

The optimal clutch size revisited: separating individual quality from the parental survival costs of reproduction

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Abstract

Life-history theory, central to our understanding of diversity in morphology, behaviour and senescence, describes how traits evolve through the optimisation of trade-offs in investment. Despite considerable study, there is only minimal support for trade-offs within species between the two traits most closely linked to fitness – reproductive effort and survival – questioning the theory's general validity. We used a meta-analysis to separate the effects of individual quality (positive survival/reproduction correlation) from the costs of reproduction (negative survival/reproduction correlation) using studies of reproductive effort and parental survival in birds. Experimental enlargement of brood size caused reduced parental survival. However, the effect size of brood size manipulation was small and opposite to the effect of phenotypic quality, as we found that individuals that naturally produced larger clutches also survived better. The opposite effects on parental survival in experimental and observational studies of reproductive effort provides the first meta-analytic evidence for theory suggesting that quality differences mask trade-offs. Fitness projections using the overall effect size revealed that reproduction presented negligible costs, except when reproductive effort was forced beyond the level observed within species, to that seen between species. We conclude that there is little support for the most fundamental life-history trade-off, between reproductive effort and survival, operating within a population. We suggest that within species, the fitness landscape of the reproduction–survival trade-off is flat until it reaches the boundaries of the between-species fast–slow life-history continuum. Our results provide a quantitative explanation as to why the costs of reproduction are not apparent and why variation in reproductive effort persists within species.

eLife assessment

The author use an approach that is in principle **useful**, comparative meta-analysis, to contribute to our understanding of life history evolution. The advance remains limited, as both the meta-analysis and the theoretical model are **incomplete**, and proper statistical and mechanistic descriptions of the simulations are lacking. A major concern is that the interpretation does not properly take into account the effect of well-characterised complexities in the relationship between clutch size and fitness in birds.

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Main

Across taxa, we see wide variation in life-history traits, such as the number of offspring produced and time spent raising young ^{1,2,3}. The central idea in life-history theory is that resources are finite, forcing trade-offs, meaning that investment in one aspect of life requires the sacrifice of another ^{4,5,6,7}. As reproduction is one of the most resource-demanding life stages, it is expected that within-species variation in offspring production will be constrained by the cost of producing and raising young. It is thought that the fitness costs of reproduction are largely incurred as a decrement to parental survival, explaining why a species evolved a certain amount of reproductive effort, whilst also explaining the fast-slow life-history continuum between reproduction and lifespan observed across species ⁶. As reproduction and survival are the two components of life-history most closely related to fitness, this central trade-off has been the subject of much theoretical and empirical research, branching fields ranging from ecology, evolutionary biology to ageing research ^{8,9,10}.

Brood size manipulations of birds in natural conditions have provided arguably the best experimental paradigm in which to test the survival costs of reproduction, as they directly alter parental care investment. Experimental increases in brood size result in increased parental effort, suggesting that parents can typically cope with increased reproductive demands ^{11,12,13}. However, the expected increased costs of reproductive effort are not always detected and the current estimate of the cost to reproduce across studies suggests only a small and inconsistent effect ^{16,17,18,19}. The absence of a cost of reproduction (determined through experimental studies) on survival means that costs must arise elsewhere or, alternatively, that individuals may differ in quality (determined through observational studies). By quality, we refer to individuals operating at their own maximum reproductive output, determined by their phenotypic condition, local or temporal genetic adaptation, and the surrounding environment, and should manifest as a positive correlation between two fitness-associated traits (such as reproduction and survival) ^{11,19,20,21,22}. The relative importance of the trade-off between reproductive effort and survival to life-history theory and the biology of ageing ⁹, therefore remains unclear. In addition, the compelling theoretical explanation for the lack of an apparent trade-off due to the confounding effects of individual quality has not been investigated on a quantitative level ^{5,19,23,24,25}. Our study aims to separate the effects of a trade-off between reproductive effort and parental survival and the effects of individual quality by comparing studies of brood manipulations and naturally varying clutch size.

Here, we present a meta-analysis that distinguishes between individual quality effects (naturally varying clutch size) and the costs of reproduction (brood manipulation studies). To do this, we tested how parental annual survival in birds is affected by the clutch size (note brood size and clutch size are used interchangeably throughout this manuscript) they cared for in two different contexts: first in brood manipulation studies and, second, in observational studies of natural variation in clutch size. We expressed changes in survival on a per-egg basis, which, for the first time, allows for a quantitative comparison across studies and thus species. We find the effect of quality is associated with higher survival chances given an increase in reproductive effort, and that this effect is opposite but equal in magnitude to the costs of reproduction. The survival trade-off for offspring production within a population is therefore offset by differences in quality, potentially constraining the evolution of higher reproductive effort. Our analysis also uniquely allowed a quantitative across species comparison, as survival risk was expressed on a per-egg basis. We transformed the response variable, scaling for variance and mean, given that a per-egg increase in clutch size does not equate to the same proportional increase in reproductive effort for all species equally. Our findings suggest that species that generally lay smaller clutches are affected more severely by brood-size manipulations. This provides evidence that trade-offs are

only detected when an individual is forced to perform far outside its optimum level (Figure S1). Species who on average have smaller clutch sizes receive disproportionate manipulations (Figure S1A) and this pushes species beyond their natural range (Figure S1B).

To predict the evolutionary consequences of the effect sizes that we estimated using the meta-analysis, we projected the fitness consequences for a change in clutch-size life-history strategy. We found that the costs to parental survival as estimated from brood size manipulation studies translate into negligible fitness costs, with a relatively flat fitness landscape, suggesting that birds underproduce in terms of clutch size, given the absence of fitness costs. This conclusion fits with our comparative finding that suggests that only brood sizes manipulated beyond the natural range incur substantial survival costs. Our results therefore suggest that, in wild populations, parental survival costs are, at most, a small component of the total fitness costs of investing parental effort. Our results do suggest that a cost of reproduction can be detected when an organism is pushed to the extreme of its reproductive capacity. We therefore infer that, though the survival–parental care trade-off does exist within species, it not strong enough to constrain clutch size and can therefore not explain clutch size evolution and thus variation within species. In addition, our work shows that differences in individual quality counterbalance the trade-off between survival and reproduction, as previously theorised [5](#), and as such could constrain reproductive effort further, and contribute to the maintenance of clutch size variation in a population.

Results

The relationship between clutch size and survival was significantly different and opposite between observational and brood manipulation studies, irrespective of how clutch size was scaled ($p < 0.01$, **Figure 1**, **Table 1**). Within natural variation (observational studies), parents with larger clutches showed increased survival (or parents with smaller clutches showed reduced survival). In contrast, when broods were experimentally manipulated, the opposite relationship was found; increasing brood sizes decreased survival. Although the difference in overall effect size between experimental and natural variation in clutch size was strongly significant in each comparison made, the individual overall effect sizes only became significant (from zero) when clutch size was expressed as a proportional increase. Expressing clutch size as a proportional increase corrects for the variation in average clutch size observed across the species included in this analysis, which ranged from 2 to 11. The parental effort required to raise two instead of one chick is potentially doubled, whereas one additional chick in a brood of 11 is likely to require only a marginal increase in effort. Indeed, also when using a between-species comparison, the effects of clutch size manipulation and quality were strongest in the species that laid the smallest clutches, suggesting that costs to survival were only observed when a species was pushed beyond its natural limits (**Figure 2**, Figure S1A, Table S1).

Males and females did not differ in their survival response to changing clutch size (Table S2, Figure S2, contrary to Santos & Nakagawa 2012 [18](#)). The variance assigned to the random effects in the model was largely accounted for by study (Table S3). Species accounted for more variation than the phylogeny, indicating that species vary in the relationship between survival and reproductive effort, irrespective of their shared evolutionary ancestry. However, our dataset included few closely related species, which reduces our ability to estimate phylogenetic effects (**Figure 3**).

Projected fitness consequences of the costs of reproduction

From our meta-analysis we now have a quantifiable and comparable effect size for the survival costs of reproduction that we can use to predict the evolutionary consequences of individuals in a given population increasing their clutch size across a range of life histories. To this end, we

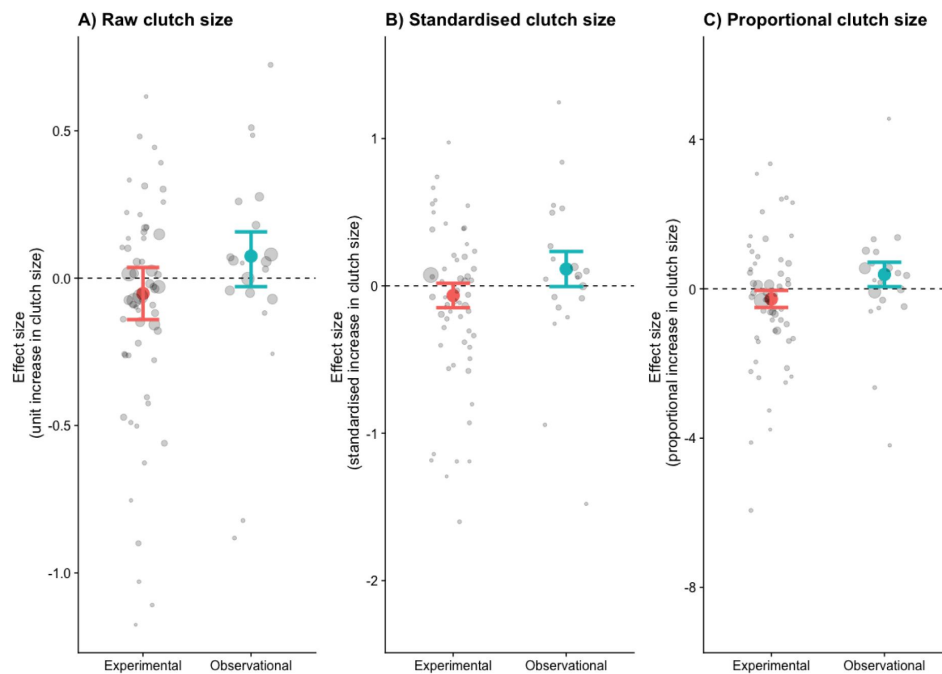


Figure 1

The effects size (log odds of survival given an increase in clutch size) for three different measures of clutch size: (A) raw, (B) standardised, and (C) proportional clutch size. Coloured points are the combined effect sizes of the odds ratios with their 95% confidence intervals. Points are coloured by whether they represent brood manipulation experiments (costs of reproduction) or they are observational (quality). Grey points are the odds ratios of each study, with their sizes weighted by the sampling variance.

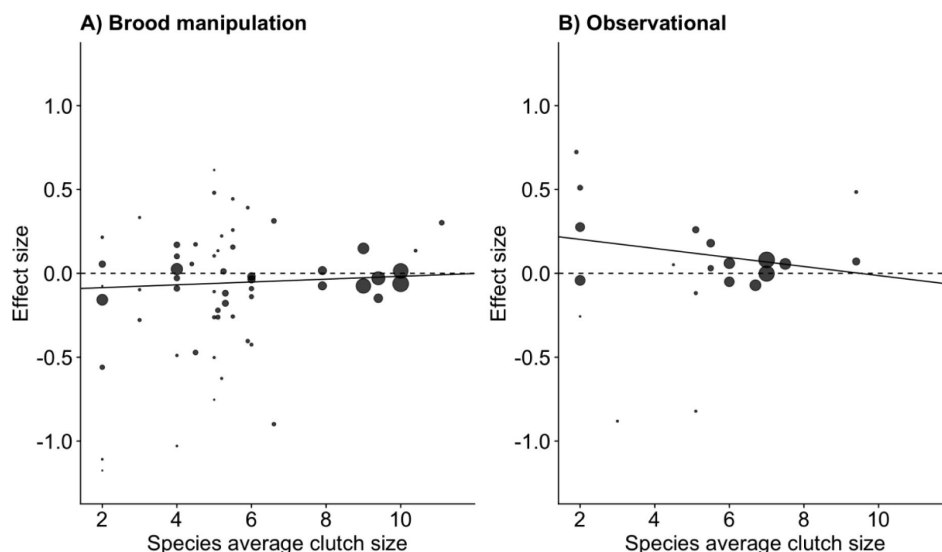


Figure 2

The meta-analytic linear regression (Table S1) of the effect size for increasing clutch size (per egg) on parental survival, given the average clutch size for the species. Species with small clutch sizes showed stronger costs of reproduction and a stronger relationship with quality (Interaction between treatment and species clutch size effect size = -0.036, $p = 0.015$). The points are the survival effect sizes (log odds ratio) per egg (as in [Figure 1A](#)) on parental survival in each study, with the point size reflecting the meta-analytic weight of that study.

Parameter			Effect	95% size	95% CI	<i>p</i>	(individual)
				lower bounds	upper bounds		
Raw	Clutch size	Brood manipulation	-0.0522	-0.1406	0.0363	0.0007	(0.2477)
		Observational	0.0747	0.1571	-0.0288		(0.1571)
Standardised	Clutch size	Brood manipulation	-0.0651	-0.1478	0.0177	0.0065	(0.1232)
		Observational	0.1143	-0.0046	0.2333		(0.0595)
Proportional	Clutch size	Brood manipulation	-0.2703	-0.4984	-0.0423	0.0005	(0.0202)
		Observational	0.3850	0.0583	0.7116		(0.0209)
Model = ~observational_or_experimental, random = (species, phylogeny, study reference)							

Table 1

Effect size estimates for the odds of survival with increasing clutch size (raw, standardised and proportional clutch size). The p-values indicate the difference between brood manipulations and observational data, with the individual effect p-values (from zero) in parentheses.

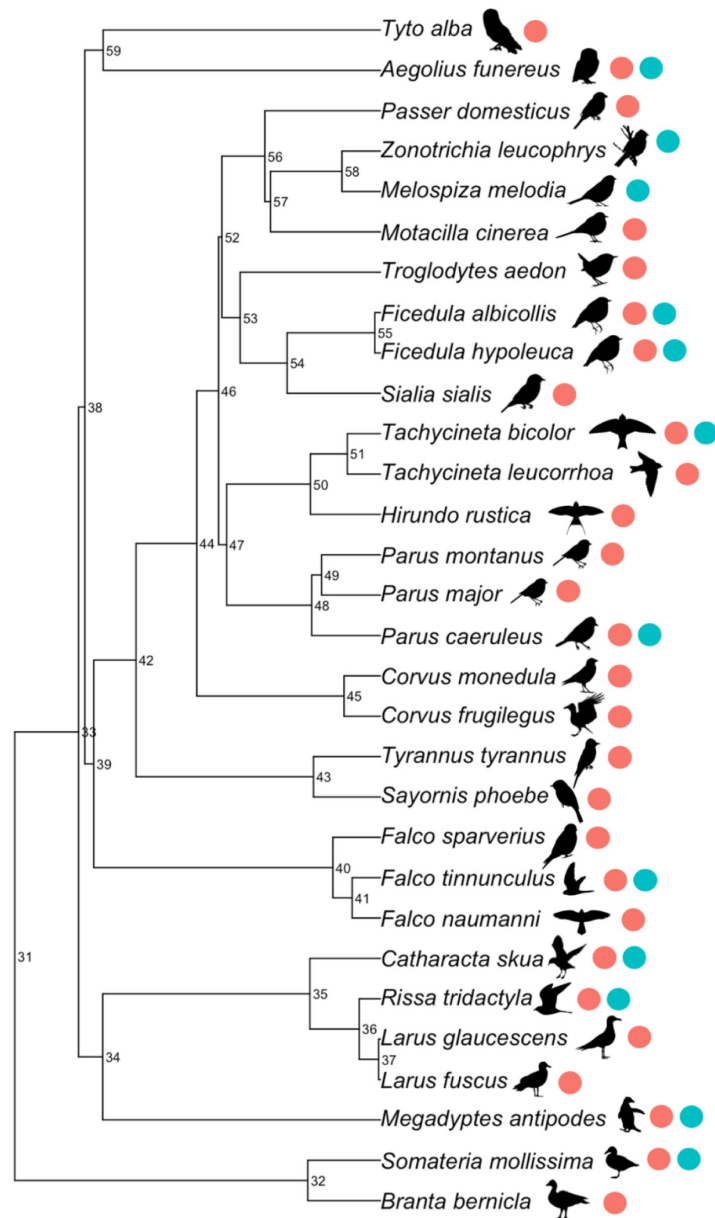


Figure 3

Phylogenetic tree of species included in our meta-analysis. Numeric labels denote the node support (as a percentage). Coloured points indicate whether the species were included in the experimental or observational studies.

projected the fitness consequences of increased reproductive effort, starting with the average effect size estimate per egg (-0.05, [Table 1](#)). Across a range of life histories, for a range of annual survival rates and clutch sizes ([Figure 4](#)). Overall, the effect size estimated in the meta-analysis (-0.05) resulted in a gain of fitness when reproductive output increased, especially in hypothetical species with low survival and small clutches. If we used a more severe effect size (-0.15, the lower confidence interval of our meta-analysis), this led to a reduction in fitness in almost all cases. Conversely, the benefit of higher reproductive output was largely offset by the cost of survival when a species' survival rate and clutch size were high. When we increased the effect size up to five-fold, fitness costs of reproduction became more pronounced, but were still not present in species with small clutches and short lifespans.

Under long-term evolution these selection differentials should lead to individual hypothetical species moving towards the diagonal (bottom left to the right top corner). This diagonal represents the observed fast-slow pace of life continuum observed among species [25](#). Exemplar species (i.e., with survival and average clutch size combinations observable in wild populations), for which we predicted the fitness consequences of the costs of reproduction, lie on this comparative diagonal in life history. In these exemplar species, the selection differential was observed to lie slightly above one, indicating that individuals having a higher clutch size than the species' average would gain a slight fitness benefit. The fitness costs and benefits did, in general, not diverge substantially with the addition of chicks, but flattened, suggesting that the costs of survival counterbalance the benefits of reproduction across a range of reproductive outputs within a species.

The low costs of reproduction that we estimated could still be responsible for between-species life-history evolution, constraining species reproductive output and survival combination to fall along the diagonal of the fast-slow pace of life continuum. How selection pressures translate into short term and longer-term evolutionary trajectories is uncertain.

Often directional selection estimated in the wild does not translate to the inter-generational change on the population level [26](#). Note, however, that only far away from the diagonal did our fitness projections reach a magnitude that would be predicted to lead to rapid evolutionary change [27](#) and see SI). The weak selection effects that lie on the diagonal are probably to be counterbalanced in the wild by factors such as environmental effects and genetic effects (e.g., gene flow from immigration or random mutation) [28](#). We argue that within-species the minimal costs of reproduction, a flat fitness landscape and quality effects ([Figure 1](#)) together explain why individuals appear to under-produce. Only when individuals are pushed beyond the observed between-species constraint do costs become apparent ([Figures 1](#), [2](#)).

Our interpretation of the reproduction/lifespan life-history trade-off, based on our quantitative meta-analysis and subsequent fitness projections, explains several key observations and contradictions in the field. A strong trade-off is observed between species, but within species this trade-off is not apparent and variation in reproductive output is maintained within fitness boundaries similar to those that determine the between-species life history trade-off. The implication of this conclusion is that the costs of reproduction are likely to operate on a physiological level, but that the fitness consequences will remain largely flat over a species' observed variation in reproductive output. These effects are further obscured by the effects of quality (observational studies), which are opposite in sign and magnitude to the cost of reproduction (experimental studies) ([Figure 1](#)) and are likely to further flatten the fitness landscape.

Selection differentials (i.e., the difference in lifetime reproductive output between hypothetical control and brood-manipulated populations) above 1 represents high lifetime fitness. Survival rates, clutch sizes, the magnitude of the manipulation (chicks added) and effect sizes represent the range of these variables present in the studies used in our meta-analysis. For each clutch size, we used a predicted survival rate and effect size to give isoclines that are biologically meaningful

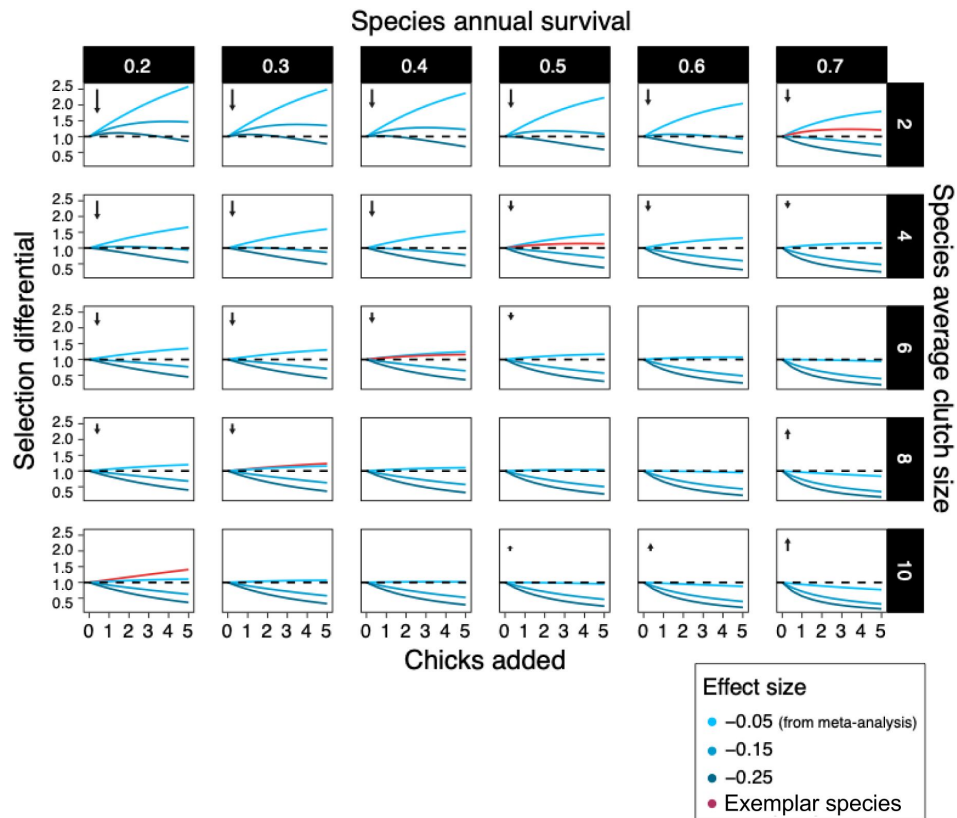


Figure 4

Isoclines of selection differentials among hypothetical control populations (in which individuals reproduce at the species' mean rate) and hypothetical brood-manipulated populations (where individuals reproduce at an increased rate compared to control) for their whole lives.

(exemplar birds shown in red). Arrows indicate the relative size and direction of selection in life-history space (on the reproduction axis). The costs of reproduction we estimated within species are predicted to result in a fast–slow life-history continuum across species, and the exemplar species we used as examples fit on this diagonal of survival rate/ clutch size combinations. We suggest that individual species show limited costs of reproduction, as they operate within relatively wide constraints imposed by the cost of reproduction that is responsible for the strong life-history trade-off observed across species.

Discussion

Our results provide the first meta-analytic evidence that differences in individual quality drive variation in clutch size and that parental survival costs do not constrain within-species variation in reproductive effort (**Figure 5**). Here, we use the definition of quality as a combination of traits that give an individual higher fitness. The finding of individual quality being a driver of variation in clutch size is not necessarily a surprising one, even though this contradicts the general theory that trade-offs drive variation in clutch size, as among-individual heterogeneity is also a well-established effect across populations in the field of ecology and evolution (such as age-specific reproductive performance). Furthermore, a recent meta-analysis on genetic life-history trade-offs found several positive genetic correlations between survival and other life-history traits, which is in conflict with life-history trade-off theory. Brood manipulations often fail to take into account that variation in resource acquisition may be larger than variation in resource allocation, and though intuitively manipulation of effort tests variation in resource allocation, it fails to account for the baseline variation in reproductive performance. This presents a fundamental flaw which our results demonstrate – variation in resource allocation can be present and testable through experiments (i.e., brood manipulations) but may only account for a small proportion of variation observed in reproductive output. Perhaps fitting with this interpretation or with additional costs from recurring bouts of enhanced reproductive effort, Jackdaws (*Corvus monedula*) show costs of increased brood size only when applied sequentially across years, but note that for this study there is no control group that is only manipulated in single years.

The reason selection is not acting on “high quality” individuals is currently unknown, but it is likely that environmental variability leads to alternative phenotypes being selected for at different points in space and time (also discussed in Pujol *et al.*, 2018). It is also possible that the quality effect could represent a terminal effect, where individuals have lower reproductive output in the year preceding their death, thereby driving the trend for naturally lower laying birds to have lower survival when estimated between individuals (e.g., Coulson & Fairweather, 2001; Rattiste, 2004; Simons *et al.*, 2016; also see Hammers *et al.*, 2012 for age-related changes in reproductive output). The effect on parental annual survival of having naturally larger clutches was significantly opposite to the result of increasing clutch size through brood manipulation, and quantitatively similar. Parents with naturally larger clutches are thus expected to live longer and this counterbalances the “cost of reproduction” when their brood size is experimentally manipulated. It is, therefore, possible that quality effects mask trade-offs. Furthermore, it could be possible that individuals that lay larger clutches have smaller costs of reproduction, i.e. would respond less in terms of annual survival to a brood size manipulation, but with our current dataset we cannot address this hypothesis (**Figure 5**). The effect of a change in clutch size on parental survival may also be non-linear, but it remains to be determined what shape, if not linear, the relationship would be. Although these possible non-linear effects warrants investigation, these relationships likely differ between species and inclusion in our current work here could lead to spurious relationships being reported.

For both costs of reproduction and quality effects, we found that species that laid the smallest clutches showed the largest effects. Brood manipulations that affect parental survival are thus likely to be the result of pushing parental effort beyond its natural optima (Figure S1) and so

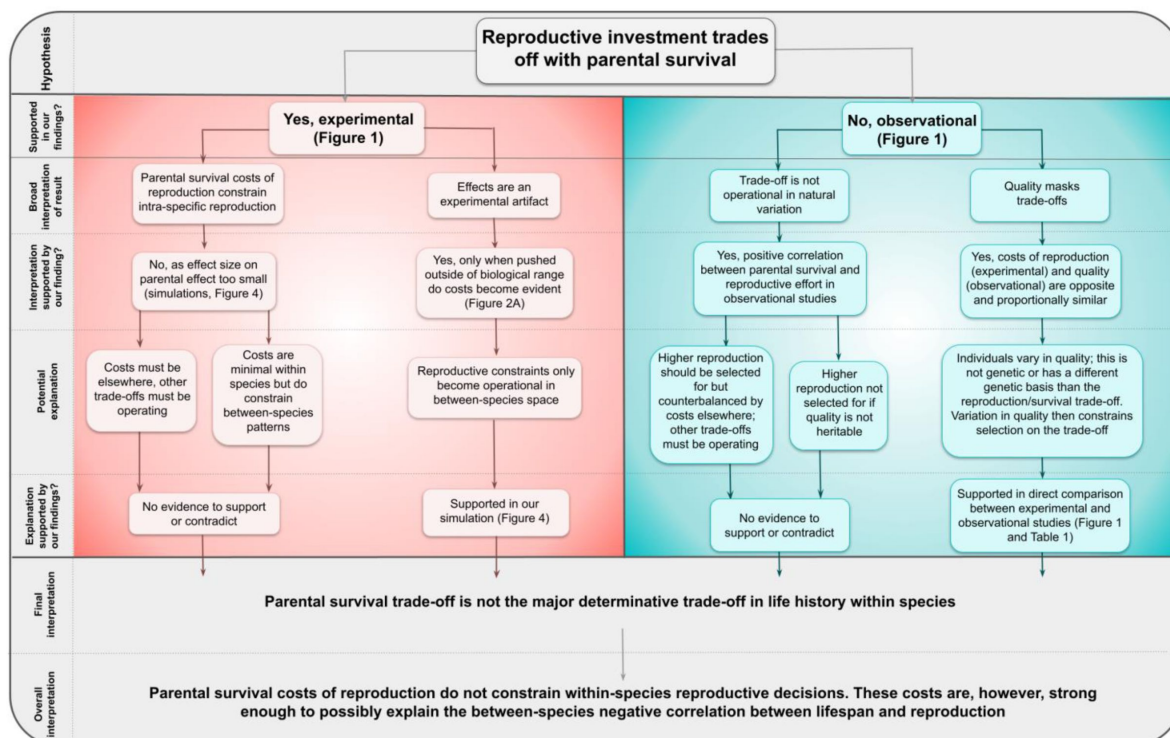


Figure 5

Decision tree representing the logical steps from our original hypothesis to our overall interpretation of our findings.

arguably that brood manipulations are not necessarily a good test of whether trade-offs happen in the wild. The classic trade-off between adult survival and the clutch size cared for is only apparent when an individual is forced to raise a clutch outside of its individual optimum, and these effects are confounded or even fully counterbalanced by differences in quality (as theorised in van Noordwijk & de Jong, 1986 ⁵).

Our fitness projections of the consequences of the costs of reproduction using the overall effect size we estimated suggest that, for current extant species, the within-species fitness landscape of the reproduction survival trade-off is flat. Species' life history decisions are constrained within a broader fast–slow life-history continuum, explaining why variation within species in reproductive effort, such as in clutch size, is large and near universal. Of course, this analysis does not fully explore more complex variation in clutch size observed in the wild, such as that observed between temperate and tropical reproductive strategies. Such work would provide useful in understanding why variation in clutch size is maintained across species with different life-history strategies. Our results do, however, provide a key step in understanding the generalised relationship between reproductive effort and survival across species (**Figure 5**). Our interpretation also assumes that other fitness costs of reproduction are smaller or at least less relevant than survival costs. However, it is possible that such costs are actually more important, and it should be noted that effects such as those on offspring quality (e.g., Conrad & Robertson, 1992 and Smith *et al.*, 1989 ^{14,35}), parental condition other than survival (e.g., Reid, 1987 and Kalmbach *et al.*, 2004 ^{36,37}) and future reproductive effort (e.g., Järvisjö *et al.*, 2016 ³⁸) have been observed (also see **Figure 5**). However, the importance of these effects is likely to vary considerably in different species. Using offspring quality as an example, some species produce sacrificial offspring, others experience catch-up growth, meaning that though the effect of increasing offspring number on offspring quality is important for certain species, drawing generalised across-species conclusions is unlikely to be possible. Conversely, the survival cost to reproduce is thought to be universal across both bird species and across taxa, and it is perhaps for this reason that the reproduction-survival trade-off has been considered to contribute more to variation in reproductive effort than other trade-offs. Our work refutes this idea directly. Indeed, the few studies that have measured the different domains that contribute to fitness in brood-size manipulation studies concluded that only in combination do these costs result in balancing selection for the current most common brood size in the population ^{39–41}. Such classic trade-off explanations do, however, fail to explain why variation in reproductive effort is prevalent within species and why between-species life-history trade-offs appear so much stronger and conclusive. Our analysis and interpretation suggest that, at its optimum, the within-species trade-off between survival and reproduction is relatively flat, and thus neutral to selection (supporting the theory presented in Cohen *et al.*, 2020 ¹⁹). We suggest that the lack of evidence supporting trade-offs driving within-species variation does not necessarily mean that physiological costs of reproduction are non-existent (e.g., Smith *et al.*, 1989 and Lemaître *et al.*, 2015 ^{35,42}), but rather that, within the wild and within the natural range of reproductive activities, such costs are not relevant to fitness. One key explanation for this effect supported by our meta-analysis (**Figure 5**) and prior theory ⁵, is that individuals differ in quality.

Methods

All data and code used in this study can be found at: <https://doi.org/10.5061/dryad.q83bk3jnk>.

Study sourcing & inclusion criteria

We extracted studies of parental survival to the following year given clutch size raised using the following inclusion criteria (similar to Santos & Nakagawa, 2012 ¹⁸): the study must be on a wild population; the study must contain variation in the number of offspring produced/raised (hereafter referred to as clutch size for simplicity), the study must report variation in clutch size in

relation to parental survival to the following year (including both experimental and observational studies) and must provide sample sizes. We did include studies where parental survival was reported for both parents combined as opposed to Santos & Nakagawa (2012) who required male and female data to be reported separately. Excluded studies and the grounds for their removal are given in the supplementary information (Table S4). We started by, first, extracting data (clutch size raised after manipulation and associated parental survival to the next year) from the included brood-manipulation studies from Santos & Nakagawa (2012) and then searched the literature to include more recently-published studies (Supplementary Methods), and in this search also added studies Santos & Nakagawa missed. In addition to brood manipulation studies, we extracted data from studies that correlated variation in parental survival with natural variation in clutch size (observational studies). For each species used in the brood manipulation studies, we aimed to find the same species for the observational studies to ensure that the effects of quality were estimated across a similar range of species. The reason for this is to ensure the experimental and observational datasets are comparable in terms of species, which could bias results if not considered, and so facilitate a more direct comparison between the quality effect (observational studies) and the trade-off (experimental studies). Where there was no equivalent study in the same species, we attempted to find a study of a congener. In most cases, observational data were obtained from either the same paper as the one describing brood manipulations (11 studies), or via searching for other papers by the same authors (2 studies). If this failed to produce observational data, a search was conducted following the same protocol as for the brood-manipulation experiments, but also specifying species, genus and/or common name in the search (7 studies). Of the 28 species used in the brood manipulation studies, we were able to find 10 of the same species in observational studies. We were able to find a further 6 species which were congeners for observational studies.

From the literature search, 78 individual effect sizes from 30 species and 46 papers were used (20 observational and 58 experimental studies). While extracting these studies we also made note of the average clutch size of the species and the within-species standard deviation in clutch size. We extracted this information from the paper containing the study but if the information was missing, we searched other published literature with the aim to find the information from a similar population (i.e., at a similar latitude).

Extracting effect sizes

We used the absolute value of offspring in the nest (for experimental studies this is the number of offspring after the manipulation occurred) and associated parental survival to estimate an effect size by performing a logistic regression to obtain the log odds ratio for parental survival, given the clutch size (i.e., positive effect sizes indicate an increased chance of survival). For example, a bird who laid a clutch size of 5 but was manipulated to have -1 chick was recorded to have a clutch size of 4. If the manipulation was reported but absolute value of offspring produced was not, we used clutch size to be the species average $\pm$ the number of offspring added or removed. Parental survival was modelled as a binary response variable for the number of birds who survived and who died after raising a given clutch size. Clutch size was averaged (mean) if a single estimate of survival was reported for multiple clutch sizes. 'Year' was included as an explanatory variable to correct for between-year variation in adult survival, where data were presented for multiple years. We standardised the clutch size (by the mean of the species and by the within-species standard deviation in clutch size) and transformed clutch size to a proportion of the species mean. For species that have no within-species variance in clutch size, we used a value of 0.01 for the standard deviation in clutch size to prevent issues in calculations when using zero. We, therefore, expressed variation in clutch size in three ways: a raw increase in clutch size, a standardised clutch size and a proportional clutch size.

Meta-analysis

We ran a single model using the log odds ratio calculated for each clutch size transformation (i.e., raw, standardised and proportional) to determine the effect of parental survival, given an increase in reproductive effort using the *metafor* package ⁴³ in R 3.3.2 ⁴⁴. From these models we were also able to directly compare the effect size of brood manipulation studies and observational studies by including a categorical fixed effect for study type (i.e., experimental or observational). We included phylogeny as a correlation matrix in these meta-analytic models to correct for shared ancestry. The phylogeny was obtained using BEAST to measure a distribution of 1,000 possible phylogenetic trees of the focal 30 species extracted from BirdTree ⁴⁵. We also included species and each studies' journal reference as random effects in the model. From these models, we calculated the proportion of variance explained by the phylogenetic effect ⁴⁶.

We then tested the effect of the species' mean clutch size on the relationship between parental survival and clutch size. We ran a single model with the mean centred (from all species used in the meta-analysis) species' average clutch size in interaction with treatment (brood manipulation or observational). Species, phylogeny and reference were also included as random effects to correct for the similarity of effect sizes within species and studies.

The difference in survival for the different sexes was modelled for each clutch size measure. Brood manipulation studies and observational studies were analysed in separate models. Sex was modelled as a categorical moderator (41 female studies, 27 male studies and 10 mixed studies). Species, phylogeny and reference were included as random effects (Table S2 and Figure S2).

Publication bias

Much of the data used in this analysis were taken from studies where these data were not the main focus of the study. This reduces the risk that our results are heavily influenced by a publication bias for positive results. A funnel plot for the survival against raw clutch size model is presented in Supplementary Information (Figure S3).

Fitness projections

We calculated various isoclines using the brood-manipulation overall effect size (-0.05, based on raw clutch size) that we estimated. Here, an isocline is a trendline representing the change in fitness returns over an individual's lifespan, given an increase in individual clutch size. An estimated "lifetime reproductive output" was calculated for hypothetical control populations (starting with 100 individuals), where all individuals consistently reproduce at the level of a species mean and have a consistent annual survival rate. We calculated this "lifetime reproductive output" (see Supplementary Information for equation details for calculating lifetime reproductive output) using combinations of a range of species average clutch sizes at 2, 4, 6, 8 and 10 and survival rates of 0.2, 0.3, 0.4, 0.5, 0.6 and 0.7, which reflected the range of clutch sizes and survival rates seen in the species in our meta-analysis. We started with a hypothetical population size of 100 and calculated the populations lifetime reproductive output rate that depends on how many offspring are produced in a clutch and how long individuals in the population live for. We could then use to model the effects of the costs of reproduction using the meta-analytic estimate of how brood size is associated with parental survival (through experimental studies).

The "lifetime reproductive output" estimate was then repeated for a hypothetical population that reproduces at an increased level compared to control, i.e., brood size enlargement, throughout their lives. These individuals therefore produce more offspring but face reductions to survival. This analysis allows us to determine if an overall fitness gain is achieved by producing more offspring despite paying an increased survival cost or whether the survival cost balances out the fitness gains of producing more offspring through reduced reproductive attempts. To obtain this,

we added a range of 1–5 offspring to the clutch sizes of the control populations. Using a range of increased clutch sizes allowed us to investigate how increased reproductive effort would affect lifetime fitness. The survival costs were determined by the overall effect size found for brood manipulation studies (per egg). We modelled effect sizes of -0.05, -0.15 and -0.25, which represent, respectively, the meta-analytic overall effect size, its upper confidence interval and a further severe effect within the observed effect sizes (rounded to the closest 0.05 for simplicity). For example, an individual who has an additional offspring in its nest would see a 5%, 15% and 25% (respectively for each effect size) reduction in its survival odds compared to if it reproduced at its normal rate (i.e., the control population rate).

We then calculated the selection differential ($LRS_{\text{brood manipulation}} / LRS_{\text{control}}$) between the hypothetical control and “brood manipulation” populations for each combination of survival rate, clutch size and effect size, and plotted this as an isocline. We further plotted the fitness consequences for five exemplar species, where survival rate and clutch size combinations are observable in the wild. We used effect sizes from model predictions at these survival rate and clutch size combinations rather than the meta-analytic mean, thereby providing a biological context.

Data Availability

Data and R code used for the statistical analysis in this study are available in the Dryad repository: <https://doi.org/10.5061/dryad.q83bk3jnk> .

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Reviewer #2 (Public Review):

I have read the re-submission of the manuscript "The optimal clutch size revisited: separating individual quality from the parental survival costs of reproduction" by LA Winder and colleagues.

I have to say that I am quite disappointed not to see any formalisation of the mechanism that the authors have in mind to explain the results they have and to draw general conclusions from it. In my original review, I strongly recommended "improving the theoretical component of the analysis by providing a solid theoretical framework before, from it, drawing conclusions. This, at a minimum, requires [...] most importantly a mechanistic model describing the assumed relationships."

Without it, it is impossible to follow, agree or disagree with the authors and learn something from the meta-analysis other than: the clutch size-annual survival relationship has opposite slopes for manipulated and natural populations. Such a set of equations (would replace pages of verbose and) is not only necessary for the readers to be able to understand the authors' points and to clearly understand the simplifying assumptions, but also for the authors to ensure they conclusions are sound. For these reasons this is a central part of such studies, see, e.g. (Walker et al., 2008). This is supposedly replaced here by a figure (figure 5), which top-left part reads: "Parental survival costs of reproduction constrain intra-specific reproduction" - "no the effect size on fig 4 is too small". Figure 4 is the output of simulations where the authors have incorporated the mean effect on survival rate per egg from the manipulated populations into a model where they compute R_0 for various increases in the annual fertility rate, and related decreases in annual survival rates, showing that along the slow-fast gradient, for balanced survival-reproduction (certainly not far from $R_0=1$), R_0 is not affected (or very little) by change in fertility-survival along the trade-off. Nowhere on this figure, do we have any information inferring that survival costs of reproduction do not constrain intra-specific reproduction. It is actually possible to build a simple mechanistic model with a trade-off mechanism that strongly affects the LRS and its variance between individuals and to would produce the exact same figure.

This is compounded in this manuscript by the constant verbose, imprecisions, outright mistakes, with a general confusion between magnitudes and variation of magnitudes, which makes it very hard to read. Let us just look at two examples illustrating my points. In the abstract, I read: "... revealed that reproduction presented negligible costs, except when reproductive effort was forced beyond the level observed within species, to that seen between species" means nothing: what is the level of reproductive effort seen between species? I suppose the authors mean "forced beyond the maximum level observed within species, to that seen between species" or something like that. Caption figure 4: "Selection differentials (i.e., the difference in lifetime reproductive output between hypothetical control and brood-manipulated populations)" It cannot be how this was calculated however: the difference between equal things is 0, not 1. These errors and all the other imprecisions, lengthy definitions that are for some almost impossible to fathom are the direct result of trying at all costs not to use a single equation, the most important tool in the study of ecology and trade-offs in particular, in a paper on costs of reproduction.

<https://doi.org/10.7554/eLife.87018.2.sa0>

Author response:

The following is the authors' response to the original reviews.

eLife assessment

In this potentially useful study, the authors attempt to use comparative meta-analysis to advance our understanding of life history evolution. Unfortunately, both the meta-analysis and the theoretical model is inadequate and proper statistical and mechanistic descriptions of the simulations are lacking. Specifically, the interpretation overlooks the effect of well-characterised complexities in the relationship between clutch size and fitness in birds.

We would like to thank the reviewers for their helpful comments, which have been considered carefully and have been valuable in progressing our manuscript. The following bullet points summarise the key points and our responses, though our detailed responses to specific comments can be found below:

- Two reviewers commented that our data was not made available. Our data **was** provided upon submission and during the review process, however was not made accessible to the reviewers. Our data and code are available at <https://doi.org/10.5061/dryad.q83bk3jnk>.
- The reviewers have highlighted that some of our methodology was unclear and we have added all the requested detail to ensure our methods can be easily understood.
- The reviewers highlight the importance of our conclusions, but also suggest some interpretations might be missing and/or are incomplete. To make clear how we objectively interpreted our data and the wider consequences for life-history theory we provide a decision tree (Figure 5). This figure makes clear where we think the boundaries are in our interpretation and how multiple lines of evidence converge to the same conclusions.

Reviewer #1 (Public Review):

This paper falls in a long tradition of studies on the costs of reproduction in birds and its contribution to understanding individual variation in life histories. Unfortunately, the meta-analyses only confirm what we know already, and the simulations based on the outcome of the meta-analysis have shortcomings that prevent the inferences on optimal clutch size, in contrast to the claims made in the paper.

There was no information that I could find on the effect sizes used in the meta-analyses other than a figure listing the species included. In fact, there is more information on studies that were not included. This made it impossible to evaluate the data-set. This is a serious omission, because it is not uncommon for there to be serious errors in meta-analysis data sets. Moreover, in the long run the main contribution of a meta-analysis is to build a data set that can be included in further studies.

It is disappointing that two referees comment on data availability, as we supplied a link to our full dataset and the code we used in Dryad with our submitted manuscript. We were also asked to supply our data during the review process and we again supplied a link to our dataset and code, along with a folder containing the data and code itself. We received confirmation that the reviewers had been given our data and code. We support open science and it was our intention that our dataset should be fully available to reviewers and readers. Our data and code are at <https://doi.org/10.5061/dryad.q83bk3jnk>.

The main finding of the meta-analysis of the brood size manipulation studies is that the survival costs of enlarging brood size are modest, as previously reported by Santos & Nakagawa on what I suspect to be mostly the same data set.

Public Reviews:

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The main finding of the meta-analysis of the brood size manipulation studies is that the survival costs of enlarging brood size are modest, as previously reported by Santos & Nakagawa on what I suspect to be mostly the same data set.

We disagree that the main finding of our paper is the small survival cost of manipulated brood size. The major finding of the paper, in our opinion, is that the effect sizes for experimental and observational studies are in opposite directions, therefore providing the first quantitative evidence to support the influential theoretical framework put forward by van Noordwijk and de Jong (1986), that individuals differ in their optimal clutch size and are constrained to reproducing at this level due to a trade-off with survival. We further show that while the manipulation experiments have been widely accepted to be informative, they are not in fact an effective test of whether within-species variation in clutch size is the result of a trade-off between reproduction and survival.

The comment that we are reporting the same finding as Santos & Nakagawa (2012) is a misrepresentation of both that study and our own. Santos & Nakagawa found an effect of parental effort on survival **only** in males who had their clutch size increased – but no effect for males who had their clutch size reduced and no survival effect on females for either increasing or reducing parental effort. However, we found an overall reduction in survival for birds who had brood sizes manipulated to be larger than their original brood (for both sexes and mixed sex studies combined). In our supplementary information, we demonstrate that the overall survival effect of a change in reproductive effort is close to zero for males, negative (though non-significant) for females and significantly negative for mixed sexes (which are not included in the Santos & Nakagawa study). Please also note that the Santos & Nakagawa study was conducted over 10 years ago. This means we added additional data (L364-365). Furthermore, meta-analyses are an evolving practice and we also corrected and improved on the overall analysis approach (e.g. L358-359 and L 393-397, and see detailed SI).

The paper does a very poor job of critically discussing whether we should take this at face value or whether instead there may be short-comings in the general experimental approach. A major reason why survival cost estimates are barely significantly different from zero may well be that parents do not fully adjust their parental effort to the manipulated brood size, either because of time/energy constraints, because it is too costly and therefore not optimal, or because parents do not register increased offspring needs. Whatever the reason, as a consequence, there is usually a strong effect of brood size manipulation on offspring growth and thereby presumably their fitness prospects. In the simulations (Fig.4), the consequences of the survival costs of reproduction for optimal clutch size were investigated without considering brood size manipulation effects on the offspring. Effects on offspring are briefly acknowledged in the discussion, but otherwise ignored. Assuming that the survival costs of reproduction are indeed difficult to discern because the offspring bear the brunt of the increase in brood size, a simulation that ignores the latter effect is unlikely to yield any insight in optimal clutch size. It is not clear therefore what we learn from these calculations.

The reviewer's comment is somewhat of a paradox. We take the best studied example of the trade-off between reproductive effort and parental survival – a key theme in life history and the biology of ageing – and subject this to a meta-analysis. The reviewer suggests we should interpret our finding as if there must be something wrong with the method or studies we included, rather than considering that the original hypothesis could be false or inflated in importance. We do not consider questioning the premise of the data over questioning a favoured hypothesis to necessarily be the best scientific approach here. In many places in our manuscript, we question and address, at length, the underlying data and their interpretation (L116-117, L165-167, 202-204 and L277-282). Moreover, we make it clear that we focus on the trade-off between current reproductive effort and subsequent parental survival, while being aware that other trade-offs could counter-balance or explain our findings (discussed on L208-210 & L301-316). Note that it is also problematic, when you do not find the expected response, to search for an alternative that has not been measured. In the case here, of potential trade-offs, there are endless possibilities of where a trade-off might operate between traits. We purposefully focus on the one well-studied and most commonly invoked trade-off. We clearly acknowledge, though, that when all possible trade-offs are taken into account a trade-off on the fitness level can occur and cite two famous studies (Daan et al., 1990 and Verhulst & Tinbergen 1991) that have shown just that (L314-316).

So whilst we agree with the reviewer that the offspring may incur costs themselves, rather than costs being incurred by the parents, the aim of our study was to test for a general trend across species in the survival costs of reproductive effort. It is unrealistic to suggest that incorporating offspring growth into our simulations would add insight, as a change in offspring number rarely affects all offspring in the nest equally and there can even be quite

stark differences; for example, this will be most evident in species that produce sacrificial offspring. This effect will be further confounded by catch-up growth, for example, and so it is likely that increased sibling competition from added chicks alters offspring growth trajectories, rather than absolute growth as the reviewer suggests. There are mixed results in the literature on the effect of altering clutch size on offspring survival, with an increased clutch size through manipulation often increasing the number of recruits from a nest.

What we do appreciate from the reviewer's comment is that the interpretation of our findings is complex. Even though our in-text explanation includes the caveats the reviewer refers to, and are discussed at length, their inter-relationships are hard to appreciate from a text format. To improve this presentation and for ease of the reader, we have added a decision tree (Figure 5) which represents the logical flow from the hypothesis being tested through to what overall conclusion can be drawn from our results. We believe this clarifies what conclusions can be drawn from our results. We emphasise again that the theory that trade-offs between reproductive effort and parental survival being the major driver of variation in offspring production was not supported though is the one that practitioners in the field would be most likely to invoke, and our result is important for this reason.

There are other reasons why brood size manipulations may not reveal the costs of reproduction animals would incur when opting for a larger brood size than they produced spontaneously themselves. Firstly, the manipulations do not affect the effort incurred in laying eggs (which also biases your comparison with natural variation in clutch size). Secondly, the studies by Boonekamp et al on Jackdaws found that while there was no effect of brood size manipulation on parental survival after one year of manipulation, there was a strong effect when the same individuals were manipulated in the same direction in multiple years. This could be taken to mean that costs are not immediate but delayed, explaining why single year manipulations generally show little effect on survival. It would also mean that most estimates of the fitness costs of manipulated brood size are not fit for purpose, because typically restricted to survival over a single year.

First, our results did show a survival cost of reproduction for brood manipulations (L107-123, Figure 1, Table 1). Note, however, that much theory is built on the immediate costs of reproduction and, as such, these costs are likely overinterpreted, meaning that our overall interpretation still holds, i.e. "parental survival trade-off is not the major determinative trade-off in life history within-species" (Figure 5).

We agree with the reviewer that lifetime manipulations could be even more informative than single-year manipulations. Unfortunately, there are currently too few studies available to be able to draw generalisable conclusions across species for lifetime manipulations. This is, however, the reason we used lifetime change in clutch size in our fitness projections, which the reviewer seems to have missed – please see methods line 466-468, where we explicitly state that this is lifetime enlargement. Of course, such interpretations do not include an accumulation of costs that is greater than the annual cost, but currently there is no clear evidence that such an assumption is valid. Such a conclusion can also not be drawn from the study on jackdaws by Boonekamp et al (2014) as the treatments were life-long and, therefore, cannot separate annual from accrued (multiplicative) costs that are more than the sum of the annual costs incurred. Note that we have now included specific discussion of this study in response to the reviewer (L265-269).

Details of how the analyses were carried out were opaque in places, but as I understood the analysis of the brood size manipulation studies, manipulation was coded as a covariate, with negative values for brood size reductions and positive values for brood size enlargements (and then variably scaled or not to control brood or clutch size). This approach implicitly assumes that the trade-off between current brood size

(manipulation) and parental survival is linear, which contrasts with the general expectation that this trade-off is not linear. This assumption reduces the value of the analysis, and contrasts with the approach of Santos & Nakagawa.

We thank the reviewer for highlighting a lack of clarity in places in our methods. We have added additional detail to the methodology section (see “Study sourcing & inclusion criteria” and “Extracting effect sizes”) in our revised manuscript. Note, that our data and code was not shared with the reviewers despite us supplying this upon submission and again during the review process, which would have explained a lot more of the detail required.

For clarity in our response, each effect size was extracted by performing a logistic regression with survival as a binary response variable and clutch size was the absolute value of offspring in the nest (i.e., for a bird that laid a clutch size of 5 but was manipulated to have -1 egg, we used a clutch size value of 4). The clutch size was also standardised and, separately, expressed as a proportion of the species’ mean.

We disagree that our approach reduces the value of our analysis. First, our approach allows a direct comparison between experimental and observational studies, which is the novelty of our study. Our approach does differ from Santos & Nakagawa but we disagree that it contrasts. Our approach allows us to take into consideration the severity of the change in clutch size, which Santos & Nakagawa do not. Therefore, we do not agree that our approach is worse at accounting for non-linearity of trade-offs than the approach used by Santos & Nakagawa. Arguably, the approach by Santos & Nakagawa is worse, as they dichotomise effort as increased or decreased, factorise their output and thereby inflate their number of outcomes, of which only 1 cell of 4 categories is significant (for males and females, increased and decreased brood size). The proof is in the pudding as well, as our results clearly demonstrate that the magnitude of the manipulation is a key factor driving the results, i.e. one offspring for a seabird is a larger proportion of care (and fitness) than one offspring for a passerine. Such insights were not achieved by Santos & Nakagawa’s method and, again, did not allow a direct quantitative comparison between quality (correlational) and experimental (brood size manipulation, i.e. “trade-off”) effects, which forms a central part of our argumentation (Figure 5).

Our analysis, alongside a plethora of other ecological studies, does assume that the response to our predictor variable is linear. However, it is common knowledge that there are very few (if any) truly linear relationships. We use linear relationships because they serve a good approximation of the trend and provide a more rigorous test for an underlying relationship than would fitting nonlinear models. For many datasets the range of added chicks required to estimate a non-linear relationship was not available. The question also remains of what the shape of such a non-linear relationship should be and is hard to determine a priori. There is also a real risk when fitting non-linear terms that they are spurious and overinterpreted, as they often present a better fit (denoting one df is not sufficient especially when slopes vary). We have added this detail to our discussion.

The observational study selection is not complete and apparently no attempt was made to make it complete. This is a missed opportunity - it would be interesting to learn more about interspecific variation in the association between natural variation in clutch size and parental survival.

We clearly state in our manuscript that we deliberately tailored the selection of studies to match the manipulation studies (L367-369). We paired species extracted for observational studies with those extracted in experimental studies to facilitate a direct comparison between observational and experimental studies, and to ensure that the respective datasets were comparable. The reviewer’s focus in this review seems to be solely on the experimental dataset. This comment dismisses the equally important observational component of our

analysis and thereby fails to acknowledge one of the key questions being addressed in this study. Note that in our revised version we have edited the phylogenetic tree to indicate for which species we have both types of information, which highlights our approach to selecting observational data (Figure 3).

Reviewer #2 (Public Review):

I have read with great interest the manuscript entitled "The optimal clutch size revisited: separating individual quality from the costs of reproduction" by LA Winder and colleagues. The paper consists in a meta-analysis comparing survival rates from studies providing clutch sizes of species that are unmanipulated and from studies where the clutch sizes are manipulated, in order to better understand the effects of differences in individual quality and of the costs of reproduction. I find the idea of the manuscript very interesting. However, I am not sure the methodology used allows to reach the conclusions provided by the authors (mainly that there is no cost of reproduction, and that the entire variation in clutch size among individuals of a population is driven by "individual quality").

We would like to highlight that we do **not** conclude that there is no cost of reproduction. Please see lines 336–339, where we state that our lack of evidence for trade-offs driving within-species variation in clutch size does not necessarily mean the costs of reproduction are non-existent. We conclude that individuals are constrained to their optima by the survival cost of reproduction. It is also an over-statement of our conclusion to say that we believe that variation in clutch size is only driven by quality. Our results show that unmanipulated birds that have larger clutch sizes also lived longer, and we suggest that this is evidence that some individuals are “better” than others, but we do not say, nor imply, that no other factors affect variation in clutch size. We have added Figure 5 to our manuscript to help the reader better understand what questions we can answer with our study and what conclusions we can draw from our results.

I write that I am not sure, because in its current form, the manuscript does not contain a single equation, making it impossible to assess. It would need at least a set of mathematical descriptions for the statistical analysis and for the mechanistic model that the authors infer from it.

We appreciate this comment, and have explained our methods in terms that are accessible to a wider audience. Note, however, that our meta-analysis is standard and based on logistic regression and standard meta-analytic practices. We have added the model formula to the model output tables.

For the simulation, we simply simulated the resulting effects. We of course supplied our code for this along with our manuscript (<https://doi.org/10.5061/dryad.q83bk3jnk>), though as we mentioned above, we believe this was not shared with the reviewers despite us making this available for the review process. We therefore understand why the reviewer feels the simulations were not explained thoroughly. We have revised our methods section and added details which we believe make our methodology more clear without needing to consult the supplemental material. However, we have also added the equations used in the process of calculating our simulated data to the Supplementary Information for readers who wish to have this information in equation form.

The texts mixes concepts of individual vs population statistics, of within individual vs among-individuals measures, of allocation trade-offs and fitness trade-offs, etcwhich means it would also require a glossary of the definitions the authors use for these various terms, in order to be evaluated.

We would like to thank the reviewer for highlighting this lack of clarity in our text. Throughout the manuscript we have refined our terminology and indicated where we are referring to the individual level or the population level. The inclusion of our new Figure 5 (decision tree) should also help in this context, as it is clear on which level we base our interpretation and conclusions on.

This problem is emphasised by the following sentence to be found in the discussion "The effect of birds having naturally larger clutches was significantly opposite to the result of increasing clutch size through brood manipulation". The "effect" is defined as the survival rate (see Fig 1). While it is relatively easy to intuitively understand what the "effect" is for the unmanipulated studies: the sensitivity of survival to clutch size at the population level, this should be mentioned and detailed in a formula. Moreover, the concept of effect size is not at all obvious for the manipulated ones (effect of the manipulation? or survival rate whatever the manipulation (then how could it measure a trade-off ?)? at the population level? at the individual level ?) despite a whole appendix dedicated to it. This absolutely needs to be described properly in the manuscript.

Thank you for identifying this sentence for which the writing was ambiguous, our apologies. We have now rewritten this and included additional explanation. L282-290: 'The effect on parental annual survival of having naturally larger clutches was significantly opposite to the result of increasing clutch size through brood manipulation, and quantitatively similar. Parents with naturally larger clutches are thus expected to live longer and this counterbalances the "cost of reproduction" when their brood size is experimentally manipulated. It is, therefore, possible that quality effects mask trade-offs. Furthermore, it could be possible that individuals that lay larger clutches have smaller costs of reproduction, i.e. would respond less in terms of annual survival to a brood size manipulation, but with our current dataset we cannot address this hypothesis (Figure 5).'

We would also like to thank the reviewer for bringing to our attention the lack of clarity about the details of our methodology. We have added details to our methodology (see "Extracting effect sizes" section) to address this (see highlighted sections). For clarity, the effect size for both manipulated and unmanipulated nests was survival, given the brood size raised. We performed a logistic regression with survival as a binary response variable (i.e., number of individuals that survived and number of individuals that died after each breeding season), and clutch size was the absolute value of offspring in the nest (i.e., for a bird that laid a clutch size of 5 but was manipulated to have -1 egg, we used a clutch size value of 4). This allows for direct comparison of the effect size (survival given clutch size raised) between manipulated and unmanipulated birds.

Despite the lack of information about the underlying mechanistic model tested and the statistical model used, my impression is still that the interpretation in the introduction and discussion is not granted by the outputs of the figures and tables. Let's use a model similar to that of (van Noordwijk and de Jong, 1986): imagine that the mechanism at the population level is

$$a.c_{(i,q)} + b.s_{(i,q)} = E_q$$

Where $c_{(i,q)}$ are $s_{(i,q)}$ are respectively the clutch size for individual i which is of quality q , and E_q is the level of "energy" that an individual of quality q has available during the given time-step (and a and b are constants turning the clutch size and survival rate into energy cost of reproduction and energy cost of survival, and there are both quite "high" so that an extra egg ($c_{(i,q)}$ is increased by 1) at the current time-step, decreases $s_{(i,q)}$ markedly (E_q is independent of the number of eggs produced), that is, we have strong individual costs of reproduction). Imagine now that the variance of $c_{(i,q)}$ (when the

population is not manipulated) among individuals of the same quality group, is very small (and therefore the variance of $s_{(i,q)}$ is very small also) and that the expectation of both are proportional to E_q . Then, in the unmanipulated population, the variance in clutch size is mainly due to the variance in quality. And therefore, the larger the clutch size $c_{(i,q)}$ the higher E_q , and the higher the survival $s_{(i,q)}$.

In the manipulated populations however, because of the large a and b , an artificial increase in clutch size, for a given E_q , will lead to a lower survival $s_{(i,q)}$. And the "effect size" at the population level may vary according to a, b and the variances mentioned above. In other words, the costs of reproduction may be strong, but be hidden by the data, when there is variance in quality; however there are actually strong costs of reproduction (so strong actually that they are deterministic and that the probability to survive is a direct function of the number of eggs produced)

We would like to thank the reviewer for these comments. We have added detail to our methodology section so our models and rationale are more clear. Please note that our simulations only take the experimental effect of brood size on parental survival into account. Our model does not incorporate quality effects. The reviewer is right that the relationship between quality and the effects exposed by manipulating brood size can take many forms and this is a very interesting topic, but not one we aimed to tackle in our manuscript. In terms of quality we make two points: (1) overall quality effects connecting reproduction and parental survival are present, (2) these effects are opposite in direction to the effects when reproduction is manipulated and similar in magnitude. We do not go further than that in interpreting our results. The reviewer is correct, however, that we do suggest and repeat suggestions by others that quality can also mask the trade-off in some individuals or circumstances (L74-76, L95-98 & L286-289), but we do not quantify this, as it is dependent on the unknown relationship between quality and the response to the manipulation. A focussed set of experiments in that context would be interesting and there are some data that could get at this, i.e. the relationship between produced clutch size and the relative effect of the manipulation (now included L287-290). Such information is, however, not available for all studies and, although we explored the possibility of analysing this, currently this is not possible with adequate confidence and there is the possible complexity of non-linear effects. We have added this rationale in our revision (L259-265).

Moreover, it seems to me that the costs of reproduction are a concept closely related to generation time. Looking beyond the individual allocative (and other individual components of the trade-off) cost of reproduction and towards a populational negative relationship between survival and reproduction, we have to consider the intra-population slow fast continuum (some types of individuals survive more and reproduce less (are slower) than other (which are faster)). This continuum is associated with a metric: the generation time. Some individuals will produce more eggs and survive less in a given time-period because this time-period corresponds to a higher ratio of their generation time (Gaillard and Yoccoz, 2003; Gaillard et al., 2005). It seems therefore important to me, to control for generation time and in general to account for the time-step used for each population studied when analysing costs of reproduction. The data used in this manuscript is not just clutch size and survival rates, but clutch size per year (or another time step) and annual (or other) survival rates.

The reviewer is right that this is interesting. There is a longstanding unexplained difference in temperate (seasonal) and tropical reproductive strategies. Most of our data come from seasonal breeders, however. Although there is some variation in second brooding and such, these species mostly only produce one brood. We do agree that a wider consideration here is relevant, but we are not trying to explain all of life history in our paper. It is clearly the case that other factors will operate and the opportunity for trade-offs will vary among species

according to their respective life histories. However, our study focuses on the two most fundamental components of fitness – longevity and reproduction – to test a major hypothesis in the field, and we uncover new relationships that contrast with previous influential studies and cast doubt on previous conclusions. We question the assumed trade-off between reproduction and annual survival. We show that quality is important and that the effect we find in experimental studies is so small that it can only explain between-species patterns but is unlikely to be the selective force that constrains reproduction within species. We do agree that there is a lot more work that can be done in this area. We hope we are contributing to the field, by questioning this central trade-off. We have incorporated some of the reviewers suggestions in the revision (L309-315). We have added Figure 5 to make clear where we are able to reach solid conclusions and the evidence on which these are based as clearly as possible in an easily accessible format.

Finally, it is important to relate any study of the costs of reproduction in a context of individual heterogeneity (in quality for instance), to the general problem of the detection of effects of individual differences on survival (see, e.g., Fay et al., 2021). Without an understanding of the very particular statistical behaviour of survival, associated to an event that by definition occurs only once per life history trajectory (by contrast to many other traits, even demographic, where the corresponding event (production of eggs for reproduction, for example) can be measured several times for a given individual during its life history trajectory).

Thank you for raising this point. The reviewer is right that heterogeneity can dampen or augment selection. Note that by estimating the effect of quality here we give an example of how heterogeneity can possibly do exactly this. We thank the reviewer for raising that we should possibly link this to wider effects of heterogeneity and we have added to our discussion of how our results play into the importance of accounting for among-individual heterogeneity (L252-256).

References:

- Fay, R. et al. (2021) 'Quantifying fixed individual heterogeneity in demographic parameters: Performance of correlated random effects for Bernoulli variables', *Methods in Ecology and Evolution*, 2021(August), pp. 1-14. doi: 10.1111/2041-210x.13728.
- Gaillard, J.-M. et al. (2005) 'Generation time: a reliable metric to measure life-history variation among mammalian populations.', *The American naturalist*, 166(1), pp. 119-123; discussion 124-128. doi: 10.1086/430330.
- Gaillard, J.-M. and Yoccoz, N. G. (2003) 'Temporal Variation in Survival of Mammals: a Case of Environmental Canalization?', *Ecology*, 84(12), pp. 3294-3306. doi: 10.1890/02-0409.
- van Noordwijk, A. J. and de Jong, G. (1986) 'Acquisition and Allocation of Resources: Their Influence on Variation in Life History Tactics', *American Naturalist*, p. 137. doi: 10.1086/284547.

Reviewer #3 (Public Review):

The authors present here a comparative meta-analysis analysis designed to detect evidence for a reproduction/ survival trade-off, central to expectations from life history theory. They present variation in clutch size within species as an observation in conflict with expectations of optimisation of clutch size and suggest that this may be accounted for from weak selection on clutch size. The results of their analyses support this explanation - they found little evidence of a reproduction - survival trade-off across birds. They extrapolated from this result to show in a mathematical model that the fitness consequences of enlarged clutch sizes would only be expected to have a significant effect

on fitness in extreme cases, outside of normal species' clutch size ranges. Given the centrality of the reproduction-survival trade-off, the authors suggest that this result should encourage us to take a more cautious approach to applying concepts the trade-off in life history theory and optimisation in behavioural ecology more generally. While many of the findings are interesting, I don't think the argument for a major re-think of life history theory and the role of trade-offs in fitness maximisation is justified.

The interest of the paper, for me, comes from highlighting the complexities of the link between clutch size and fitness, and the challenges facing biologists who want to detect evidence for life history trade-offs. Their results highlight apparently contradictory results from observational and experimental studies on the reproduction-survival trade-off and show that species with smaller clutch sizes are under stronger selection to limit clutch size.

Unfortunately, the authors interpret the failure to detect a life history trade-off as evidence that there isn't one. The construction of a mathematical model based on this interpretation serves to give this possible conclusion perhaps more weight than is merited on the basis of the results, of this necessarily quite simple, meta-analysis. There are several potential complicating factors that could explain the lack of detection of a trade-off in these studies, which are mentioned and dismissed as unimportant (lines 248-250) without any helpful, rigorous discussion. I list below just a selection of complexities which perhaps deserve more careful consideration by the authors to help readers understand the implications of their results:

We would like to thank the reviewer for their thoughtful response and summary of the findings that we also agree are central to our study. The reviewer also highlights areas where our manuscript could benefit from a deeper consideration and we have added detail accordingly to our revised discussion.

We would like to highlight that we do not interpret the failure to detect a trade-off as evidence that there is not one. First, and importantly, we **do** find a trade-off but show this is only incurred when individuals produce a clutch beyond their optimal level. Second, we also state on lines 322-326 that the lack of evidence to support trade-offs being strong enough to drive variation in clutch size does not necessarily mean there are no costs of reproduction.

The statement that we have constructed a mathematical model based on the interpretation that we have not found a trade-off is, again, factually incorrect. We ran these simulations because the opposite is true – we did find a trade-off. There is a significant effect of clutch size when manipulated on annual parental survival. We benefit from our unique analysis allowing for a quantitative fitness estimate from the effect size on annual survival (as this is expressed on a per-egg basis). This allowed us to ask whether this quantitative effect size can alone explain why reproduction is constrained, and we evaluate this using simulations. From these simulations we find that this effect size is too small to explain the constraint, so something else must be going on, and we do spend a considerable amount of text discussing the possible explanations (L202-215). Note that the possibly most parsimonious conclusion here is that costs of reproduction are not there, or simply small, so we also give that explanation some thought (L221-224 and L315-331).

We are disappointed by the suggestion that we have dismissed complicating factors that could prevent detection of a trade-off, as this was not our intention. We were aiming to highlight that what we have demonstrated to be an apparent trade-off can be explained through other mechanisms, and that the trade-off between clutch size and survival is not as strong in driving within-species variation in clutch size as previously assumed. We have added further discussion to our revised manuscript to make this clear and give readers a better understanding of the complexity of factors associated with life-history theory, including the addition of a decision tree (Figure 5).

• *Reproductive output is optimised for lifetime reproductive success and so the consequences of being pushed off the optimum for one breeding attempt are not necessarily detectable in survival but in future reproductive success (and, therefore, lifetime reproductive success).*

We agree this is a valid point, which is mentioned in our manuscript in terms of alternative stages where the costs of reproduction might be manifested (L316-320). We would also like to highlight that, in our simulations, the change in clutch size (and subsequent survival cost) was assumed for the lifetime of the individual, for this very reason.

• *The analyses include some species that hatch broods simultaneously and some that hatch sequentially (although this information is not explicitly provided (see below)). This is potentially relevant because species which have been favoured by selection to set up a size asymmetry among their broods often don't even try to raise their whole broods but only feed the biggest chicks until they are sated; any added chicks face a high probability of starvation. The first point this observation raises is that the expectation of more chicks = more cost, doesn't hold for all species. The second more general point is that the very existence of the sequential hatching strategy to produce size asymmetry in a brood is very difficult to explain if you reject the notion of a trade-off.*

We agree with the reviewer that the costs of reproduction can be absorbed by the offspring themselves, and may not be equal across offspring (we also highlight this at L317-318 in the manuscript). However, we disagree that for some species the addition of more chicks does not equate to an increase in cost, though we do accept this might be less for some species. This is, however, difficult to incorporate into a sensible model as the impacts will vary among species and some species do also exhibit catch-up growth. So, without a priori knowledge on this, we kept our model simple to test whether the effect on parental survival (often assumed to be a strong cost) can explain the constraint on reproductive effort, and we conclude that it does not.

We would also like to make clear that we are not rejecting the notion of a trade-off. Our study shows evidence that a trade-off between survival and reproductive effort probably does not drive within-species variation in clutch size. We do explicitly say this throughout our manuscript, and also provide suggestions of other areas where a trade-off may exist (L317-320). The point of our study is not whether trade-offs exist or not, it is whether there is a generalisable across-species trend for a trade-off between reproductive effort and survival – the most fundamental trade-off in our field but for which there is a lack of conclusive evidence within species. We believe the addition of Figure 5 to our reviewed manuscript also makes this more evident.

• *For your standard, pair-breeding passerine, there is an expectation that costs of raising chicks will increase linearly with clutch size. Each chick requires X feeding visits to reach the required fledge weight. But this is not the case for species which lay precocious chicks which are relatively independent and able to feed themselves straight after hatching - so again the relationship of care and survival is unlikely to be detectable by looking at the effect of clutch size but again, it doesn't mean there isn't a trade-off between breeding and survival.*

Precocial birds still provide a level of parental care, such as protection from predators. Though we agree that the level of parental care in provisioning food (and in some cases in all parental care given) is lower in precocial than altricial birds, this would only make our reported effect size for manipulated birds to be an underestimate. Again, we would like to draw the reviewer's attention to the fact we did detect a trade-off in manipulated birds and

we do not suggest that trade-offs do not exist. The argument the reviewer suggests here does not hold for unmanipulated birds, as we found that birds that naturally lay larger clutch sizes have higher survival.

• *The costs of raising a brood to adulthood for your standard pair-breeding passerine is bound to be extreme, simply by dint of the energy expenditure required. In fact, it was shown that the basal metabolic rate of breeding passerines was at the very edge of what is physiologically possible, the human equivalent being cycling the Tour de France (Nagy et al. 1990). If birds are at the very edge of what is physiologically possible, is it likely that clutch size is under weak selection?*

If birds are at the very edge of what is physiologically possible, then indeed it would necessarily follow that if they increase the resource allocated in one area then expenditure in another area must be reduced. In many studies, however, the overall brood mass is increased when chicks are added and cared for in an experimental setting, suggesting that birds are not operating at their limit all the time. Our simulations show that if individuals increase their clutch size, the survival cost of reproduction counterbalances the fitness gained by increasing clutch size and so there is no overall fitness gain to producing more offspring. Therefore, selection on clutch size is constrained to the within-species level. We do not say in our manuscript that clutch size is under weak selection – we only ask why variation in clutch size is maintained if selection always favours high-producing birds.

• *Variation in clutch size is presented by the authors as inconsistent with the assumption that birds are under selection to lay the Lack clutch. Of course, this is absurd and makes me think that I have misunderstood the authors' intended point here. At any rate, the paper would benefit from more clarity about how variable clutch size has to be before it becomes a problem for optimality in the authors' view (lines 84-85; line 246). See Perrins (1965) for an exquisite example of how beautifully great tits optimise clutch size on average, despite laying between 5-12 eggs.*

We thank the reviewer for highlighting that our manuscript may be misleading in places, however, we are unsure which part of our conclusions the author is referring to here. The question we pose is “Why don’t all birds produce a clutch size at the population optimum?”, and is central to the decades-long field of life-history theory. Why is variation maintained? As the reviewer outlines, there is extensive variability, with some birds laying half of what other birds lay.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) *Title: while the costs of reproduction are possibly important in shaping optimal clutch size, it is not clear what you can about it given that you do not consider clutch / brood size effects on fitness prospects of the offspring.*

We have expanded on our discussion of how some costs may be absorbed by the offspring themselves. However, a change in offspring number rarely affects all offspring in the nest equally and there can even be quite stark differences; for example this will be most evident in species that produce sacrificial offspring. This effect will be further confounded by catch-up growth. There are mixed results in the literature on the effect of altering clutch size on offspring survival, with an increased clutch size through manipulation often increasing the number of recruits from a nest. We have focussed on the relationship between reproductive effort and survival because it is given the most weight in the field in terms of driving intra-specific variation in clutch size. We have altered our title to show we focus on the survival

costs specifically: “The optimal clutch size revisited: separating individual quality from the parental survival costs of reproduction”.

(2) L.11-12: *I agree that this is true for birds, but this is phrased more generally here. Are you sure that that is justified?*

The trade-off between survival and reproductive effort has largely been tested experimentally through brood manipulations in birds as this provides a good system in which to test the costs and benefits of increasing parental effort. The work in this area has provided theory beyond just passerine birds, which are the most commonly manipulated group, to across-taxa theories. We are unaware of any study/studies that provide evidence that the reproduction/survival trade-off is generalisable across multiple species in any taxa. As such, we do believe this sentence is justified. An example is the lack of a consistent negative genetic correlation in populations of fruitflies, for example, that has also been hailed as a lack-of-cost paradigm. Furthermore, some mutants that live longer do so without a cost on reproduction.

(3) L.13-14: *Not sure what you mean with this sentence - too much info lacking.*

We have added some detail to this sentence.

(4) L.14: *it is slightly awkward to say 'parental investment and survival' because it is the survival effect that is usually referred to as the 'investment'. Perhaps what you want to say is 'parental effort and survival'?*

We have replaced “parental investment” with “reproductive effort”

(5) L.15: *you can omit 'caused'. Compared to control treatment or to reduced broods? Why not mention effects or lack thereof of brood reduction? And it would be good to also mention here whether effects were similar in the sexes.*

Please see our methodology where we state that we use clutch size as a continuous variable (we do not compare to control or reduced but include the absolute value of offspring in a logistic regression). The effects of a brood reduction are drawn from the same regression and so are opposite. Though we appreciate the detail here is lacking to fully comprehend our study, we would like to highlight this is the abstract and details are provided in the main text.

(6) L. 15: *I am not sure why you write 'however', as the finding that experimental and natural variation have opposite effects is in complete agreement with what is generally reported in the literature and will therefore surprise no one that is aware of the literature.*

We use “however” to highlight the change in direction of the effect size from the results in the previous sentence. We also believe that ours is the first study that provides a quantitative estimate of this effect and that previous work is largely theoretical. The reviewer states that this is what is generally reported but it is not reported in all cases, as some relationships between reproductive effort and survival are negative (for the quality measurement, in correlational space, see Figure 1).

(7) L.16: *saying 'opposite to the effect of phenotypic quality' seems difficult to justify, as clutch size cannot be equated with phenotypic quality. Perhaps simply say 'natural variation in clutch size'? If that is what you are referring to.*

Please note we are referring to effect sizes here — that is, the survival effect of a change in clutch size. By phenotypic quality we are referring to the fact that we find higher parental survival when natural clutch sizes are higher. It is not the case that we refer to quality only as having a higher clutch size. This is explicitly stated in the sentence you refer to. We have changed “effect” to “effect size” to highlight this further.

(8) L.18: why do you refer to 'parental care' here? Brood size is not equivalent to parental care.

Brood size manipulations are used to manipulate parental care. The effect on parental survival is expected to be incurred because of the increase in parental care. We have changed “parental care” to “reproductive effort” to reduce the number of terms we use in our manuscript.

(9) L.18-19: suggest to tone down this claim, as this is no more than a meta-analytic confirmation of a view that is (in my view) generally accepted in the field. That does not mean it is not useful, just that it does not constitute any new insight.

We are unaware of any other study which provides generalisable across-species evidence for opposite effects of quality and costs of reproduction. The work in this area is also largely theoretical and is yet to be supported experimentally, especially in a quantitative fashion. It is surprising to us that the reviewer considers there to be general acceptance in a field, rather than being influenced by rigorous testing of hypotheses, made possible by meta-analysis, the current gold standard in our field.

(10) L.21: what does 'parental effort' mean here? You seem to use brood size, parental care, parental effort, and parental investment interchangeably but these are different concepts. Daan et al (1990, Behaviour), which you already cite, provide a useful graph separating these concepts. Please adjust this throughout the manuscript, i.e. replace 'reproductive effort' with wording that reflect the actual variable you use.

We have not used the phrase “parental effort” in this sentence. We agree these are different concepts but in this context are intertwined. For example, brood size is used to manipulate parental care as a result of increased parental effort. We do agree the manuscript would benefit from keeping terminology consistent throughout the manuscript and have adjusted this throughout.

(11) L.23: perhaps add 'in birds' somewhere in this sentence? Some reference to the assumptions underlying this inference would also be useful. Two major assumptions being that birds adjusted their effort to the manipulation as they would have done had they opted for a larger brood size themselves, and that the costs of laying and incubating extra eggs can be ignored. And then there is the effect that laying extra eggs will usually delay the hatch date, which in many species reduces reproductive success.

Though our study does exclusively use birds, birds have been used to test the survival/reproduction trade-off because they present a convenient system in which to experimentally test this. The conclusions from these studies have a broader application than in birds alone. We believe that although these details are important, they are not appropriate in the abstract of our paper.

(12) L.26: how is this an explanation? It just repeats the finding.

We intend to refer to all interpretations from all results presented in our manuscript. We have made this more clear by adjusting our writing.

(13) L.27: *I do not see this point. And 'reproductive output' is yet another concept, that can be linked to the other concepts in the abstract in different ways, making it rather opaque.*

We have changed “reproductive output” to “reproductive effort”.

(14) L.33: *here you are jumping from 'resources' to 'energetically' - it is not clear that energy is the only or main limiting resource, so why narrow this down to energy?*

We do not say energy is the only or main limiting resource. We simply highlight that reproduction is energetically demanding and so, intuitively, a trade-off with a highly energetically demanding process would be the focal place to observe a trade off. We have, though, replaced “energetically” with “resource”.

(15) L.35-36: *this is new to me - I am not aware of any such claims, and effects on the residual reproductive value could also arise through effects on future reproduction. The authors you cite did not work on birds, or (in their own study systems) presented results that as far as I remember warrant such a general statement.*

The trade-off between reproduction and survival is seminal to the disposable soma theory, proposed by Kirkwood. Though Kirkwood's work was largely not focussed on birds, it had fundamental implications for the field of evolutionary ecology because of the generalisable nature of his proposed framework. In particular, it has had wide-reaching influence on how the biology of aging is interpreted. The readership of the journal here is broad, and our results have implications for that field too. The work of Kirkwood (many of the papers on this topic have over 2000 citations each) has been perhaps overly influential in many areas, so a link to how that work should be interpreted is highly relevant. If the reviewer is interested in this topic the following papers by one of the co-authors and others could be of interest, some of which we could not cite in the main manuscript due to space considerations:

<https://www.science.org/doi/pdf/10.1126/sciadv.aay3047>

https://agingcelljournal.org/Archive/Volume3/stochasticity_explains_non_genetic_inheritance_of_lifespan/

<https://pubmed.ncbi.nlm.nih.gov/21558242/>

<https://besjournals.onlinelibrary.wiley.com/doi/full/10.1111/1365-2435.13444>

<https://www.nature.com/articles/362305a0>

[https://www.cell.com/trends/ecology-evolution/fulltext/S0169-5347\(12\)00147-4](https://www.cell.com/trends/ecology-evolution/fulltext/S0169-5347(12)00147-4)

[https://www.cell.com/cell/pdf/S0092-8674\(15\)01488-9.pdf](https://www.cell.com/cell/pdf/S0092-8674(15)01488-9.pdf)

<https://bmcbiol.biomedcentral.com/articles/10.1186/s12915-018-0562-z>

(16) L.42: *this could be preceded with mentioning the limitations of observational data.*

We have added detail as to why brood manipulations are a good test for trade-offs and so this is now inherently implied.

(17) L.42-43: *why?*

We have added detail to this sentence.

(18) L.45: do any of the references cited here really support this statement? I am certain that several do not - in these this statement is an assumption rather than something that is demonstrated. It may be useful to look at Kate Lessell's review on this that appeared in Etologia, I think in the 1990's. Mind however that 'reproductive effort' is operationally poorly defined for reproducing birds - provisioning rate is not necessarily a good measure of effort in so far as there are fitness costs.

We have updated the references to support the sentence.

(19) L.47: Given that you make this statement with respect to brood size manipulations in birds, it seems to me that the paper by Santos & Nakagawa is the only paper you should cite here. Given that you go on to analyze the same data it deserves to be discussed in more detail, for example to clarify what you aim to add to their analysis. What warrants repeating their analysis?

Please first note that our dataset includes Santos & Nakagawa and additional studies, so it is not accurate to say we analyse the same data. Furthermore, we believe our study has implications beyond birds alone and so believe it is appropriate to cite the papers that do support our statement. We have added details to the methods to explicitly state what data is gathered from Santos & Nakagawa (it is only used to find the appropriate literature and data was re-extracted and re-analysed in a more appropriate way) and, separately, how we gathered the observational studies (see L352-381).

(20) L.48: There are more possible explanations to this, which deserve to be discussed. For example, brood size manipulations may not have been that effective in manipulating reproductive effort - for example, effects on energy expenditure tend to be not terribly convincing. Secondly, the manipulations do not affect the effort incurred in laying eggs (which also biases your comparison with natural variation in clutch size). Thirdly, the studies by Boonekamp et al on Jackdaws found that while there was no effect of brood size manipulation on parental survival after one year of manipulation, there was a strong effect when the same individuals were manipulated in the same direction in multiple years. This could be taken to mean that costs are not immediate but delayed, explaining why single year manipulations generally show little effect on survival. It would also mean that most estimates of the fitness costs of manipulated brood size are not fit for purpose, because typically restricted to survival over a single year.

Please see our response to this comment in the public reviews.

Out of interest and because the reviewer mentioned “energy expenditure” specifically: There are studies that show convincing effects of brood size manipulation on parental energy expenditure. We do agree that there are also studies that show ceilings in expenditure. We therefore disagree that they “tend to be not terribly convincing”. Just a few examples:

<https://academic.oup.com/beheco/article/10/5/598/222025> (Figure 2)

<https://besjournals.onlinelibrary.wiley.com/doi/full/10.1111/1365-2435.12321> (Figure 1)

<https://besjournals.onlinelibrary.wiley.com/doi/full/10.1046/j.1365-2656.2000.00395.x> (but ceiling at enlarged brood).

(21) L.48, "or, alternatively, that individuals may differ in quality": how do you see that happening when brood size is manipulated, and hence 'quality' of different experimental categories can be assumed to be approximately equal? This point does apply to

observational studies, so I assume that that is what you had in mind, but that distinction should be clear (also on line 54).

We have made it more clear that we determine if there are quality effects separate to the costs of reproduction found using brood manipulation studies.

(22) L.50: Drent & Daan, in their seminal paper on "The prudent parent" (1980, Ardea) were among the earliest to make this point and deserve to be cited here.

We have added this citation

(23) L.51, "relative importance": relative to what? Please be more specific.

We have adjusted this sentence.

(24) L.54: Vedder & Bouwhuis (2018, Oikos) go some way towards this point and should be explicitly mentioned with reference to the role of 'quality' effects on the association between reproductive output and survival.

We have added this reference.

(25) L.55: can you be more specific on what you want to do exactly? What you write here could be interpreted differently.

We have added an explicit aim after this sentence to be more clear.

(26) L.57: Here also a more specific wording would be useful. What does it mean exactly when you say you will distinguish between 'quality' and 'costs'?

We have added detail to this sentence.

(27) L.62: it should be clearer from the introduction that this is already well known, which will indirectly emphasize what you are adding to what we know already.

We would argue this is not well known and has only been theorised but not shown empirically, as we do here.

(28) L.62: you equate clutch size with 'quality' here - that needs to be spelled out.

We refer to quality as the positive effect size of survival for a given clutch size, not clutch size alone. We appreciate this is not clear in this sentence and have reworded.

(29) L.64: this looks like a serious misunderstanding to me, but in any case, these inferences should perhaps be left to the discussion (this also applies to later parts of this paragraph), when you have hopefully convinced readers of the claims you make on lines 62-63.

We are unsure of what the reviewer is referring to as a misunderstanding. We have chosen this format for the introduction to highlight our results. If this is a problem for the editors we will change as required.

(30) L.66: quantitative comparison of what?

Comparison of species. We have changed the wording of this sentence

(31) L.67-69: *this should be in the methods.*

We have used a modern format which highlights our result. We are happy to change the format should the editors wish us to.

(32) L.74-88: *suggest to (re)move this entire paragraph, presenting inferences in such an uncritical manner before presenting the evidence is inappropriate in my view. I have therefore refrained from commenting on this paragraph.*

We have chosen a modern format which highlights our result. We are happy to change the format should the editors wish us to.

(33) L.271, "must detail variation in the number of raised young": *it is not sufficiently clear what this means - what does 'detail' mean in this context? And what does 'number of raised young' mean? The number hatched or raised to fledging?*

We have now made this clear.

(34) L.271, "must detail variation in the number of raised young": *looking at table S4, it seems that on the basis of this criterion also brood size manipulation studies where details on the number of young manipulated were missing are excluded. I see little justification for this - surely these manipulations can for example be coded as for example having the average manipulation size in the meta-analysis data set, thereby contributing to tests of manipulation effects, but not to variation within the manipulation groups?*

We have done in part what the reviewer describes. We are specifically interested in the manipulation size, so we required this to compare effect sizes across species and categories, a key advance of our study and outlined in many places in our manuscript. Note, however, that we only need comparative differences, and have used clutch size metrics more generally to obtain a mean clutch size for a species, as well as SD where required. Please also note that our supplement details exactly why studies were excluded from our analysis, as is the preferred practice in a meta-analysis.

(35) L.271, "referred to as clutch size": *the point of this simplification is not clear to me why it is clearly confusing - why not refer to 'brood size' instead?*

Brood size and clutch size can be used interchangeably here because, in the observational studies, the individuals vary in the number of eggs produced, whereas for brood manipulations this obviously happens after hatching and brood is perhaps a more appropriate term, but we wanted to simplify the terminology used. However, we use clutch size throughout as the aim of our study is to determine why individuals differ in the number of offspring they produce, and so clutch size is the most appropriate term for that.

(36) L.280: *according to the specified inclusion criteria (lines 271/272) these studies should already be in the data set, so what does this mean exactly?*

Selection criteria refers to whether a given study should be kept for analysis or not. It does not refer to how studies were found. Please see lines 361-378 for details on how we found studies (additional details are also in the Supplementary Methods).

(37) L.281: *the use of 'quality' here is misleading - natural variation in clutch or brood size will have multiple causes, variation in phenotypic quality of the individuals and their*

environment (territories) is only one of the causes. Why not simply refer to what you are actually investigating: natural and experimental variation in brood size.

We disagree, our study aims to separate quality effects from the costs of reproduction and we use observational studies to test for quality differences, though we make no inference about the mechanisms. We do not imply that the environment causes differences in quality, but that to directly compare observation and experimental groups, they should contain similar species. So, to be clear again, quality refers to the positive covariation of clutch size with survival. We feel that we explain this clearly in our study's rationale and have also improved our writing in several sections on this to avoid any confusion (see responses to earlier comments by the three reviewers).

(38) L.283, "in most cases": please be exact and say in xx out xx cases.

We have added the number of studies for each category here.

(39) L.283-285: presumably readers can see this directly in a table with the extracted data?

Our data and code can be accessed with the following link: <https://doi.org/10.5061/dryad.q83bk3jnk>. We believe the data are too large to include as a table in the main text and are not essential in understanding the paper. Though we do believe all readers should have access to this information if they wish and so is publicly available.

(40) L.293: there does not seem to be a table that lists the included studies and effect sizes. It is not uncommon to find major errors in such tables when one is familiar with the literature, and absence of this information impedes a complete assessment of the manuscript.

We supplied a link to our full dataset and the code we used in Dryad with our submitted manuscript. We were also asked to supply our data during the review process and we again supplied a link to our dataset and code, along with a folder containing the data and code itself. We received confirmation that the reviewers had been given our data and code. We support open science and it was our intention that our dataset should be fully available to reviewers and readers. We believe the data are too large to include as a table in the main text and are not essential in understanding the paper. Our data and code are at <https://doi.org/10.5061/dryad.q83bk3jnk>.

(41) L.293: from how many species?

We have added this detail.

(42) L.296, "longevity": this is a tricky concept, not usually reported in the studies you used, so please describe in detail what data you used.

We have removed longevity as we did not use this data in our current version of the manuscript.

(43) L. 298: again: where can I see this information?

Our data and code can be accessed with the following link: <https://doi.org/10.5061/dryad.q83bk3jnk>. We did supply this information when we submitted our manuscript and again during the review process but we believe this was not passed onto the reviewers.

(44) L. 304, "we used raw data": I assume that for the majority of papers the raw data were not available, so please explain how you dealt with this. Or perhaps this applies to a selection of the studies only? Perhaps the experimental studies?

By raw data, we mean the absolute value of offspring in the nest. We have changed the wording of this sentence and added detail about whether the absolute value of offspring was not present for brood manipulation studies (L393-397).

(45) L.304: When I remember correctly, Santos and Nakagawa examined effects of reducing and enlarging brood size separately, which is of importance because trade-off curves are unlikely to be linear and whether they are or not has major effects on the optimization process. But perhaps you tackled this in another way? I will read on.....

You are correct that Santos & Nakagawa compared brood increases and reductions to control separately. Note that this only partially accounts non-linearity and it does not take into account the severity of the change in brood size. By using a logistic regression of absolute clutch size, as we have done, we are able to directly compare brood manipulations with experimental studies. Please see Supplementary Methods lines 11-12, where we have added additional detail as to why our approach is beneficial in this analysis.

(46) L.319: what are you referring to exactly with "for each clutch size transformation"?

We refer to the raw, standardised and proportional clutch size transformations. We have added detail here to be more clear.

(47) L.319: is there a cost of survival? Perhaps you mean 'survival cost'? This would be appropriate for the experimental data, but not for the observational data, where the survival variation may be causally unrelated to the brood size variation, even if there is a correlation.

We have changed "cost of survival" to "effect of parental survival". We only intend to imply causality for the experimental studies. For observational studies we do not suggest that increasing clutch size is causal for increasing survival, only correlative (and hence we use the phrase "quality").

(48) L.320: please replace "parental effort" with something like 'experimental change in brood size'.

We have changed "parental effort" to "reproductive effort"

(49) L.321: due to failure of one or more eggs to hatch, and mortality very early in life, before brood sizes are manipulated, it is not likely that say an enlargement of brood size by 1 chick can be equated to the mean clutch size +1 egg / check. For example, in the Wytham great tit study, as re-analysed by Richard Pettifor, a 'brood size manipulation' of unmanipulated birds is approximately -1, being the number of eggs / chicks lost between laying and the time of brood size manipulation. Would this affect your comparisons?

Though we agree these are important factors in determining what a clutch/brood size actually is for a given individual/pair, as this can vary from egg laying to fledging. We do not believe that accounting for this (if it was possible to do so) would significantly affect our conclusions, as observational studies are comparable in the fact that these birds would also likely see early life mortality of their offspring. It is also possibly the case that parents already

factor in this loss, and so a brood manipulation still changes the parental care effort an individual has to incur.

(50) L.332: *instead of "adjusted" perhaps say 'mean centred'?*

We have implemented this suggestion.

(51) L.345: *this statement surprised me, but is difficult to verify because I could not locate a list of the included studies. However, to my best knowledge, most studies reporting brood size manipulation effects on parental survival had this as their main focus, in contrast to your statement.*

Our data and code can be accessed with the following link: <https://doi.org/10.5061/dryad.q83bk3jnk>. We did supply this information when we submitted our manuscript and again during the review process but we believe this was not passed onto the reviewers by the journal, although supplied by us on several occasions. We regret that the reviewer was impeded by this unfortunate communication failure, but we did our best to make the data available to the reviewers during the initial review process.

(52) L.361-362: *this seems a realistic approach from an evolutionary perspective, but we know from the jackdaw study by Boonekamp that the survival effect of brood size manipulation in a single year is very different from the survival effect of manipulating as in your model, i.e. every year of an individual's life the same manipulation. For very short-lived species this possibly does not make much difference, but for somewhat longer-lived species this could perhaps strongly affect your results. This should be discussed, and perhaps also explored in your simulations?*

Note that the Boonekamp study does not separate whether the survival effects are additive or multiplicative. As such, we do not know whether the survival effects for a single year manipulation are just small and hard to detect, or whether the survival effects are multiplicative. Our simulations assumed that the brood enlargement occurred every year throughout their lives. We have added some text to the discussion on the point you raise.

(53) L.360: *what is "lifetime reproductive fitness"? Is this different from just "fitness"?*

We have changed “lifetime reproductive fitness” to “lifetime reproductive output”.

(54) L.363: *when you are interested in optimal clutch size, why not also explore effects of reducing clutch size?*

As we find that a reduction in clutch size leads to a reduction in survival (for experimental studies), we already know that these individuals would have a reduced fitness return compared to reproducing at their normal level, and so we would not learn anything from adding this into our simulations. The interest in using clutch size enlargements is to find out why an individual does not produce more offspring than it does, and the answer is that it would not have a fitness benefit (unless its clutch size and survival rate combination is out of the bounds of that observable in the wild).

(55) Fig.1 - *using 'parental effort' in the y-axis label is misleading, suggest to replace with e.g. "clutch or brood size". Using "clutch size" in the title is another issue, as the experimental studies typically changed the number of young rather than the number of eggs.*

We have updated the figure axes to say “clutch size” rather than “parental effort”. Please see response to comment 35 where we explain our use of the term “clutch size” throughout this manuscript.

(56) L.93 - 108: I appreciate the analysis in Table 1, in particular the fact that you present different ways of expressing the manipulation. However, in addition, I would like to see the results of an analysis treating the manipulations as factor, i.e. without considering the scale of the manipulation. This serves two purposes. Firstly, I believe it is in the interest of the field that you include a detailed comparison with the results of Santos & Nakagawa's analysis of what I expect to be largely the same data (manipulation studies only - for this purpose I would also like to see a comparison of effect size between the sexes). Secondly, there are (at least) two levels of meta-analysis, namely quantifying an overall effect size, and testing variables that potentially explain variation in effect size. You are here sort of combining the two levels of analysis, but including the first level also would give much more insight in the data set.

Our main intention here was to improve on how the same hypothesis was approached by Santos & Nakagawa. We did this by improving our analysis (on a by “egg” basis) and by adding additional studies (i.e. more data). In this process mistakes are corrected (as we re-extracted all data, and did not copy anything across from their dataset – which was used simply to ensure we found the same papers); more recent data were also added, including studies missed by Santos & Nakagawa. This means that the comparison with Santos & Nakagawa becomes somewhat irrelevant, apart from maybe technical reasons, i.e. pointing out mistakes or limitations in certain approaches. We would not be able to pinpoint these problems clearly without considering the whole dataset, yet Santos & Nakagawa only had a small subset of the data that were available to us. In short, meta-analysis is an iterative process and similar questions are inevitably analysed multiple times and updated. This follows basic meta-analytic concepts and Cochrane principles. Except where there is a huge flaw in a prior dataset or approach (like we sometimes found and highlighted in our own work, e.g. Simons, Koch, Verhulst 2013, Aging Cell), in itself a comparison of the kind the reviewer suggests distracts from the biology. With the dataset being made available others can make these comparisons, if required. On the sex difference, we provide a comparison of effect sizes separated between both sexes and mixed sex in Table S2 and Figure S1.

(57) L.93 - 108: a thing that does not become clear from this section is whether experimentally reducing brood size affects parental survival similarly (in absolute terms) as enlarging brood size. Whether these effects are symmetric is biologically important, for example because of its effect on clutch size optimization. In the text you are specific about the effects of increasing brood size, but the effect you find could in theory be due entirely to brood size reduction.

We have added detail to make it clear that a brood reduction is simply the opposite trend. We use linear relationships because they serve a good approximation of the trend and provide a more rigorous test for an underlying relationship than would fitting nonlinear models. For many datasets there is not a range of chicks added for which a non-linear relationship could be estimated. The question also remains of what the shape of this non-linear relationship should be and is hard to determine a priori.

We have added some discussion on this to our manuscript (L278-282), in response to an earlier comment.

(58) L.103-107: this is perhaps better deferred to the discussion, because other potential explanations should also be considered. For example, there have been studies suggesting

that small birds were provisioning their brood full time already, and hence had no scope to increase provisioning effort when brood size was experimentally increased.

We agree this is a discussion point but we believe it also provides an important context for why we ran our simulations, and so we believe this is best kept brief but in place. We agree the example you give is relevant but believe this argument is already contained in this section. See line 121-123 “...suggesting that costs to survival were only observed when a species was pushed beyond its natural limits”.

(59) L.103-107: this discussion sort of assumes that the results in Table 1 differ between the different ways that the clutch/brood size variation is expressed. Is there any statistical support for this assumption?

We are unsure of what the reviewer means here exactly. Note that in each of the clutch size transformations, experimental and observational effect sizes are significantly opposite. For the proportional clutch size transformation, experimental and observation studies are both separately significantly different from 0.

(60) L.104: at this point, I would like to have better insight into the data set. Specifically, a scatter plot showing the manipulation magnitude (raw) plotted against control brood size would be useful.

Our data and code can be accessed with the following link: <https://doi.org/10.5061/dryad.q83bk3jnk>. We did supply this information when we submitted our manuscript and again during the review process but we believe this was not passed onto the reviewers by the journal.

Thank you for this suggestion: this is a useful suggestion also to illustrate how manipulations are relatively stronger for species with smaller clutches, in line with our interpretation of the result presented in Figure 2. We have added Figure S1 which shows the strength of manipulation compared to the species average.

(61) L. 107: this seems a bold statement - surely you can test directly whether effect size becomes disproportionately stronger when manipulations are outside the natural range, for example by including this characterization as a factor in the models in Table 1.

It is hard to define exactly what the natural range is here, so it is not easy to factorise objectively, which is why we chose not to do this. However, it is clear that for species with small clutches the manipulation itself is often outside the natural range. Thank you for your suggestion to include a figure for this as it is clear manipulations are stronger in species with smaller clutches. We attribute this to species being forced outside their natural range. We consider our wording makes it clear that this is our interpretation of our findings and we therefore do not think this is a bold statement, especially as it fits with how we interpret our later simulations.

(62) Fig.3, legend: the term 'node support' does not mean much to me, please explain.

Node support is a value given in phylogenetic trees to dictate the confidence of a branch. In this case, values are given as a percentage and so can translate to how many times out of 100 the estimate of the phylogeny gives the same branching. Our values are low, as we have relatively few species in our meta-analysis.

(63) Fig.3: it would be informative when you indicate in this figure whether the species contributed to the experimental or the observational data set or both.

We have added into Fig 3 whether the species was observational, experimental or both.

(64) L.139: *the p-value refers to the interaction between species clutch size and treatment (observational vs. experimental), but it appears that no evidence is presented for the correlation being significant in either observational or experimental studies.*

We agree that our reporting of the effect size could be misinterpreted and have added detail here. The statistic provided describes the slopes are significantly different between observational and experimental, implying there are differences between the slopes of small and large clutch-laying species.

(65) L.140: *I am wondering to what extent these correlations, which are potentially interesting, are driven by the fact that species average clutch size was also used when expressing the manipulation effect. In other words, to what extent is the estimate on the Y-axis independent from the clutch size on the X-axis? Showing that the result is the same when using survival effect sizes per manipulation category would considerably improve confidence in this finding.*

We are unsure what the reviewer means by “per manipulation category”. Please also note that we have used a logistic regression to calculate our effect sizes of survival, given a unit increase in reproductive effort. So, for example, if a population contained birds that lay 2,3 or 4 eggs, provided that the number of birds which survived and died in each category did not change, if we changed the number of eggs raised to 10,11 or 12, respectively, then our effect size would be the same. In this way, our effect sizes are independent of the species’ average clutch size.

(66) L.145: *when I remember correctly, Santos & Nakagawa considered brood size reduction and enlargement separately. Can this explain the contrasting result? Please discuss.*

You are correct, in that Santos & Nakagawa compared reductions and enlargements to controls separately. However, we found some mistakes in the data extracted by Santos & Nakagawa that we believe explain the differences in our results for sex-specific effect sizes. We do not feel that highlighting these mistakes in the main text is fair, useful or scientifically relevant, as our approach is to improve the test of the hypothesis.

(67) L.158-159: *looking at table S2 it seems to me you have a whole range of estimates. In any case, there is something to be said for taking the estimates for females because it is my impression (and experience) that clutch size variation in most species is a sex-linked trait, in that clutch size tends to be repeatable among females but not among males.*

We agree that, in many cases, the female is the one that ultimately decides on the number of chicks produced. We did also consider using female effect sizes only, however, we decided against this for the following reasons: (1) many of the species used in our meta-analysis exhibit biparental care, as is the case for many seabirds, and so using females only would bias our results towards species with lower male investment; in our case this would bias the results towards passerine species. (2) it has also been shown that, as females in some species are operating at their maximum of parental care investment, it is the males who are able to adjust their workload to care for extra offspring. (3) we are ultimately looking at how many offspring the breeding adults should produce, given the effort it costs to raise them, and so even if the female chooses a clutch size completely independently of the male, it is still the effort of both parents combined that determines whether the parents gain an overall fitness

benefit from laying extra eggs. (4) some studies did not clearly specify male or female parental survival and we would not want to reduce our dataset further.

(68) L.158-168: please explain how you incorporated brood size effects on the fitness prospects of offspring, given that it is a very robust finding of brood size manipulation studies that this affects offspring growth and survival.

We would argue this is near-on impossible to incorporate into our simulations. It is unrealistic to suggest that incorporating offspring growth into our simulations would add insight, as a change in offspring number rarely affects all offspring in the nest equally and there can even be quite stark differences; for example, this will be most evident in species that produce sacrificial offspring. This effect will be further confounded by catch-up growth, for example, and so it is likely that increased sibling competition from added chicks alters offspring growth trajectories, rather than absolute growth as the reviewer suggests. There are mixed results in the literature on the effect of altering clutch size on offspring survival, with an increased clutch size through manipulation often increasing the number of recruits from a nest. It would be interesting, however, to explore this further using estimates from the literature, but this is beyond our current scope, and would in our initial intuition not be very accurate. It would be interesting to explore how big the effect on offspring should be to constrain effect size strongly. Such work would be more theoretical. The point of our simple fitness projections here is to aid interpretation of the quantitative effect size we estimated.

(69) L.163: while I can understand that you select the estimate of -0.05 for computational reasons, it has enormous confidence intervals that also include zero. This seems problematic to me. However, in the simulations, you also examined the results of selecting -0.15, which is close to the lower end of the 95% C.I., which seems worth mentioning here already.

Thank you for this suggestion. Yes, indeed, our range was chosen based on the CI, and we have now made this explicit in the manuscript.

(70) L.210: defined in this way, in my world this is not what is generally taken to be a selection differential. Is what you show not simply scaled lifetime reproductive success?

As far as we are aware, a selection differential is the relative change between a given group and the population mean, which is what we have done here. We appreciate this is a slightly unusual context in which to place this, but it is more logical to consider the individuals who produce more offspring as carrying a potential mutation for higher productivity. However, we believe that “selection differential” is the best terminology for the statistic we present. We also detail in our methodology how we calculate this. We have adjusted this sentence to be more explicit about what we mean by selection differential.

(71) L.177-180: is this not so because these parameter values are closest to the data you based your estimates on, which yielded a low estimate and hence you see that here also?

We are unsure of what exactly the reviewer means here. The effect sizes for our exemplar species were predicted from each combination of clutch size and survival rate. Note that we used a range of effect sizes, higher than that estimated in our meta-analysis, to explore a large parameter space and that these same conclusions still hold.

(72) L.191-194: these statements are problematic, because based on the assumption that an increase in brood size does not impact the fitness prospects of the offspring, and we know this assumption to be false.

Though we appreciate that some cost is often absorbed by the offspring themselves, we are unaware of any evidence that these costs are substantial and large enough to drive within-species variation in reproductive effort, though for some specific species this may be the case. However, in terms of explaining a generalisable, across-species trend, the fitness costs incurred by a reduction in offspring quality are unlikely to be significantly larger than the survival costs to reproduce. We also find it highly unlikely the cost to fitness incurred by a reduction in offspring quality is large enough to counter-balance the effect of parental quality that we find in our observational studies. We do also discuss other costs in our discussion.

(73) L.205: *here and in other places it would be useful to be more explicit on whether in your discussion you are referring to observational or experimental variation.*

We have added this detail to our manuscript. Do note that many of our conclusions are drawn by the combination of results of experimental and observational studies. We believe the addition of Figure 5 makes this more clear to the reader.

(74) L.225: *this may be true (at least, when we overlook the misuse of the word 'quality' here), but I would expect some nuance here to reflect that there is no surprise at all in this result as this pattern is generally recognized in the literature and has been the (empirical) basis for the often-repeated explanation of why experiments are required to demonstrate trade-offs. On a more quantitative level, it is worth mentioning the paper of Vedder & Bouwhuis (2017, Oikos) that essentially shows the same thing, i.e. a positive association between reproductive output and parental survival.*

We have added some discussion on this point, including adding the citation mentioned. However, we would like to highlight that our results demonstrate that brood manipulations are not necessarily a good test of trade-offs, as they fail to recognise that individuals differ in their underlying quality. Though we agree that this result should not necessarily be a surprising one, we have also not found it to be the case that differences in individual quality are accepted as the reason that intra-specific clutch size is maintained – in fact, we find that it is most commonly argued that when costs of reproduction are not identified it is concluded that the costs must be elsewhere – yet we cannot find conclusive evidence that the costs of reproduction (wherever they lie) are driving intra-specific variation in reproductive effort. Furthermore, some studies in our dataset have reported negative correlations between reproductive effort and survival (see observational studies, Figure 1).

(75) L.225-226: *perhaps present this definition when you first use the term.*

We have added more detail to where we first use and define this term to improve clarity (L57-58).

(76) L.227-228, "currently unknown": *this statement surprised me, given that there is a plethora of studies showing within-population variation in clutch size to depend on environmental conditions, in particular the rate at which food can be gathered.*

We mean to question that if an individual is “high quality”, why is it not selected for? We have rephrased, to improve clarity.

(77) L.231: *this seems no more than a special case of the environmental effect you mention above.*

We think this is a relevant special case, as it constitutes within-individual variation in reproduction that is mistaken for between-individual variation. This is a common problem in

our field, that we feel needs addressing. We only have between-individual variation here in our study on quality, and by highlighting this we show that there might not be any variation between individuals, but this could come about fully (doubtful) or partly (perhaps likely) due to terminal effects.

(78) L235-236: *but apparently depending on how experimental and natural variation was expressed? Please specify here.*

We are not sure what results the reviewer is referring to here, as we found the same effect (smaller clutch laying species are more severely affected by a change in clutch size) for both clutch size expressed as raw clutch size and standardised clutch size.

(79) L.237: *the concept of 'limits' is not very productive here, and it conflicts with the optimality approach you apply elsewhere. What you are saying here can also be interpreted as there being a non-linear relationship between brood size manipulation and parental survival, but you do not actually test for that. A way to do this would be to treat brood size reduction and enlargement separately. Trade-off curves are not generally expected to be linear, so this would also make more sense biologically than your current approach.*

We have replaced “limits” with “optima”. We believe our current approach of treating clutch size as a continuous variable, regardless of manipulation direction, is the best approach, as it allows us to directly compare with observational studies and between species that use different manipulations (now nicely illustrated by the reviewer’s suggested Figure S1). Also note that transforming clutch size to a proportion of the mean allows us to account for the severity in change in clutch size. We also do not believe that treating reductions and enlargements separately accounts for non-linearity, as either we are separating this into two linear relationships (one for enlargements and one for reductions) or we compare all enlargements/reductions to the control, as in Santos & Nakagawa 2012, which does not take into account the severity of the increase, which we would argue is worse for accounting for non-linearity. Furthermore, in the cases where the manipulation involved one offspring only, we also cannot account for non-linearity.

(80) L.239: *assuming birds are on average able to optimize their clutch size, one could argue that any manipulation, large or small, on average forces birds to raise a number of offspring that deviates from their natural optimum. At this point, it would be interesting to discuss in some detail studies with manipulation designs that included different levels of brood size reduction/enlargement.*

We agree with the reviewer that any manipulation is changing an individual’s clutch size away from its own individual optima, which we have argued also means brood manipulations are not necessarily a good test of whether a trade-off occurs in the wild (naturally), as there could be interactions with quality – we have now edited to explicitly state this (L299-300).

(81) L.242-244: *when you choose to maintain this statement, please add something along the lines of "assuming there is no trade-off between number and quality of offspring".*

As explained above, though we agree that the offspring may incur some of the cost themselves, we are not aware of any evidence suggesting this trade-off is also large enough to drive intra-specific variation in clutch size across species. Furthermore, in the context here, the trade-off between number and quality of offspring would not change our conclusion – that the fitness benefit of raising more offspring is offset by the cost on survival. We have

added detail on the costs incurred by offspring earlier in our discussion (L309-315). The addition of Figure 5 should help interpret these data.

(82) L.253: instead of reference 30 the paper by Tinbergen et al in Behaviour (1990) seems more appropriate.

We believe our current citation is relevant here but we have also added the Tinbergen et al (1990) citation.

(83) L.253-254: such trade-offs may perfectly explain variation in reproductive effort within species if we were able to estimate cost-benefit relations for individuals. In fact, reference 29 goes some way to achieve this, by explaining seasonal variation in reproductive effort.

We are unaware of any quantitative evidence that any combination of trade-offs explains intra-specific variation in reproductive effort, especially as a general across-species trend.

(84) L.255: how does one demonstrate "between species life-history trade-offs"? The 'trade-off' between reproductive rate and survival we observe between species is not necessarily causal, and hence may not really be a trade-off but due to other factors - demonstrating causality requires some form of experimental manipulation.

Between-species trade-offs are well established in the field, stemming from GC Williams' seminal paper in 1966, and for example in r/K selection theory. It is possible to move from these correlations to testing for causation, and this is happening currently by introducing transgenes (genes from other species) that promote longevity into shorter-lived species (e.g., naked-mole rat genes into mice). As yet it is unclear what the effects on reproduction are.

(85) L.256: it is quite a big claim that this is a novel suggestion. In fact, it is a general finding in evolutionary theory that fitness landscapes tend to be rather flat at equilibrium.

It is important to note here that we simulate the effect size found, and hence this is the novel suggestion, that because the resulting fitness landscape is relatively flat there is no directional selection observed. We did not intend to suggest our interpretation of flat fitness landscapes is novel. We have changed the phrasing of this sentence to avoid misinterpretation.

(86) L.259: why bring up physiological 'costs' here, given that you focus on fitness costs? Do you perhaps mean fitness costs instead of physiological costs? Furthermore, here and in the remainder of this paragraph it would be useful to be more specific on whether you are considering natural or experimental variation.

The cost of survival is a physiological cost incurred by the reduction of self-maintenance as a result of lower resource allocation. This is one arm of fitness; we feel it would be confusing here to talk about costs to fitness, as we do not assess costs to future reproduction (which formed the large part of the critique offered by the reviewer). We would like to highlight that the aim of this manuscript was to separate costs of reproduction from the effects of quality, and this is why we have observational and experimental studies in one analysis, rather than separately. Our conclusion that we have found no evidence that the survival cost to reproduce drives within-species variation in clutch size comes both from the positive correlation found in the observational studies and our negligible fitness return estimates in our simulations. We therefore, do not believe it is helpful to separate observational and

experimental conclusions throughout our manuscript, as the point is that they are inherently linked. We hope that with the addition of Figure 5 that this is more clear.

(87) L.262: The finding that naturally more productive individuals tend to also survive better one could say is by definition explained by variation in 'quality', how else would you define quality?

We agree, and hence we believe quality is a good term to describe individuals who perform highly in two different traits. Note that we also say the lack of evidence that trade-offs drive intra-specific variation in clutch size also potentially suggests an alternative theory, including intra-specific variation driven by differences in individual quality.

Supplementary information

(88) Table S1: please provide details on how the treatment was coded - this information is needed to derive the estimates of the clutch size effect for the treatments separately.

We have added this detail.

(89) Table S2: please report the number of effect sizes included in each of these models.

We have added this detail.

(90) Table S4: references are not given. Mentioning species here would be useful. For example, Ashcroft (1979) studied puffins, which lay a single egg, making me wonder what is meant when mentioning "No clutch or brood size given" as the reason for exclusion. A few more words to explain why specific studies were excluded would be useful. For example, what does "Clutch size groups too large" mean? It surprises me that studies are excluded because "No standard deviation reported for survival" - as the exact distribution is known when sample size and proportion of survivors is known.

We have updated this table for more clarity.

(91) Fig.S1: please plot different panels with the same scale (separately for observational and experimental studies). You could add the individual data points to these plots - or at least indicate the sample size for the different categories (female, male, mixed).

We have scaled all panels to have the same y axis and added sample sizes to the figure legend.

(92) Fig.S3: please provide separate plots for experimental and observational studies, as it seems entirely plausible that the risk of publication bias is larger for observational studies - in particular those that did not also include a brood size manipulation. At the same time, one can wonder what a potential publication bias among observational studies would represent, given that apparently you did not attempt to collect all studies that reported the relevant information.

We have coloured the points for experimental and observational studies. Note that a study is an independent effect size and, therefore, does not indicate whether multiple data (i.e., both experimental and observational studies) came from the same paper. As we detail in the paper and above in our reviewer responses, we searched for observational studies from species used in the experimental studies to allow direct comparison between observational and experimental datasets.

Reviewer #2 (Recommendations For The Authors):

I strongly recommend improving the theoretical component of the analysis by providing a solid theoretical framework before, from it, drawing conclusions.

This, at a minimum, requires a statistical model and most importantly a mechanistic model describing the assumed relationships.

We thank the reviewer for highlighting that our aims and methodology are unclear in places. We have added detail to our model and simulation descriptions and have improved the description of our rationale. We also feel the failure of the journal to provide code and data to the reviewers has not helped their appreciation of our methodology and use of data.

Because the field uses the same wording for different concepts and different wording for the same concept, a glossary is also necessary.

We thank the reviewer for raising this issue. During the revision of this manuscript, we have simplified our terminology or given a definition, and we believe this is sufficient for readers to understand our terminology.

Reviewer #3 (Recommendations For The Authors):

• The files containing information of data extracted from each study were not available so it has not been possible to check how any of the points raised above apply to the species included in the study. The ms should include this file on the Supp. Info as is standard good practice for a comparative analysis.

We supplied a link to our full dataset and the code we used in Dryad with our submitted manuscript. We were also asked to supply our data during the review process and we again supplied a link to our dataset and code, along with a folder containing the data and code itself. We received confirmation that the reviewers had been given our data and code. We support open science and it was our intention that our dataset should be fully available to reviewers and readers. We believe the data is too large to include as a table in the main text and is not essential in understanding the paper. Our data and code are at <https://doi.org/10.5061/dryad.q83bk3jnk>.

• For clarity, refer to "the effect size of clutch size on survival" rather than simply "effect size". Figures 1 and 2 require cross-referencing with the main text to understand the y-axis.

We have added detail to the figure legend to increase the interpretability of the figures.

• Silhouettes in Figure 3 (or photos) would help readers without ornithological expertise to understand the taxonomic range of the species included in the analyses.

We have added silhouettes into Figure 3.

• Throughout the discussion: superscripts shouldn't be treated as words in a sentence so please add authors' names where appropriate.

We have added author names and dates where required.

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